UNIT I

Flexible Power Transmission & Design of Bearings

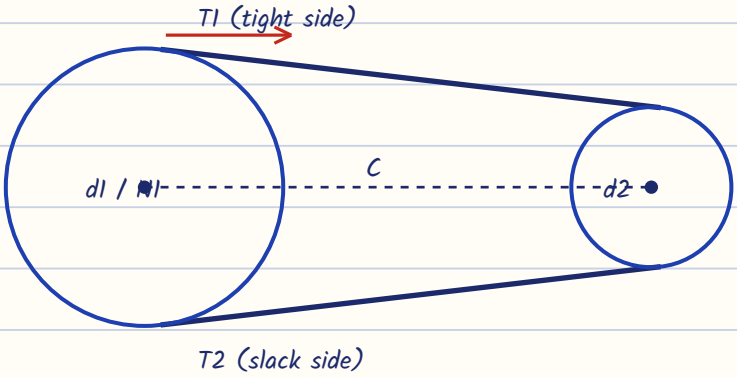
flat & V belts | pulleys | chain drives | wire ropes | lubrication | hydrodynamic theory | Sommerfeld number | McKee's equation | rolling bearings

I. Flexible power transmission – overview

Flexible drives (belts, ropes, chains) transmit power between shafts that are **far apart** where gears would be bulky. They are cheap, absorb shock and vibration, and are simple to maintain. In automobiles: **V-belt / serpentine belt** (alternator, water pump, AC compressor), **timing belt / chain** (camshaft drive), **wire ropes** (window regulators, hoists, brake cables).

Drive	Principle	Features / use
Flat belt	Friction on flat pulley	Long centre distance, high speed (up to 50 m/s), quiet; slip occurs
V-belt	Wedging in 34–40° groove	Compact, high ratio (up to 7:1), multiple belts; used in vehicles, motors
Timing (toothed) belt	Positive meshing	No slip, synchronous: cam drive, printers
Roller chain	Positive, rollers on sprocket teeth	No slip, high load, low speed, compact; bicycles, motor-cycles, timing
Wire rope	Tension member on sheave / drum	Hoists, cranes, elevators, cables over long distance

II. Belt drives – basic relations



Driver = larger pulley

Tight side is the **lower** side, slack side on top (for long drives) so wrap angle increases.

Open belt: same direction

Crossed: opposite direction

Velocity ratio, slip and creep

> **Velocity ratio** VR = N2/N1 = d1/d2 (no slip); with slip s%: N2/N1 = (d1/d2) (1 - s/100). Elastic creep adds about 0.5-1% loss.

> **Belt speed** v = π d1 N1 / 60 (m/s), usually 15-25 m/s for flat leather belts, 20-25 m/s for V-belts (max 30).

> **Creep**: relative motion because the belt stretches more on the tight side; **slip**: motion because of insufficient friction (overload).

Length of belt and angle of contact

	Open belt	**Crossed belt**
Length L	$L = \pi (r1 + r2) + 2C + (r1 - r2)^2 / C$	$L = \pi (r1 + r2) + 2C + (r1 + r2)^2 / C$
Angle of lap on small pulley θ	$\theta = 180° - 2 \sin^{-1}((r2 - r1)/C)$	$\theta = 180° + 2 \sin^{-1}((r1 + r2)/C)$
Angle on both pulleys	different (small pulley governs)	same for both pulleys

Centre distance $C \geq (d1 + d2)/2 + d1$ (roughly), and $C \geq 1.5\ d2$ for good wrap; the belt sags by about 1-2% of C.

Ratio of tensions

$$\text{Flat belt: } T1 / T2 = e^{(\mu \theta)} \qquad \text{V-belt: } T1 / T2 = e^{(\mu \theta / \sin \beta)}$$

> T1, T2 = tight & slack side tensions (N); μ = coefficient of friction; θ = angle of lap (radian); β = semi-groove angle (about 17-19° so 2β = 34-38°). The wedge makes the effective friction coefficient

$\mu/\sin\beta$ = about 3 μ - V-belt transmits much more power.

> **Power transmitted** $P = (T1 - T2)\ v$ (W); torque $T = (T1 - T2)\ d/2$.

> **Centrifugal tension** $Tc = m\ v^2$ (m = mass per metre = ρA); it lowers the useful tension: $T1 = T_{max} - Tc$, so effective ratio $(T1 - Tc)/(T2 - Tc) = e^{(\mu\theta)}$.

> **Maximum tension** $T_{max} = \sigma\ b\ t$ (allowable stress × width × thickness) ; joint efficiency 0.7-0.9 for leather (cemented 90%, laced 60-70%).

> **Initial tension** $T0 = (T1 + T2 + 2\ Tc)/2$ (= $(T1 + T2)/2$ when Tc is ignored).

> **Condition for maximum power**: $T_{max} = 3\ Tc \rightarrow v = \sqrt{(T_{max}/3m)}$.

Example 1 – flat belt power

> Belt 100 mm × 6 mm, σ_{allow} = 2 MPa, ρ = 1000 kg/m³, μ = 0.3, θ = 160° (2.79 rad), v = 15 m/s. Find power.

> $T_{max} = \sigma\ b\ t = 2 \times 100 \times 6 = 1200$ N ; $m = \rho A = 1000 \times 0.1 \times 0.006 = 0.6$ kg/m ; $Tc = m\ v^2 = 0.6 \times 225 = 135$ N.

> $T1 = 1200$ N ; $(T1 - Tc)/(T2 - Tc) = e^{(0.3 \times 2.79)} = 2.31 \rightarrow T2 = 135 + (1200 - 135)/2.31 = 596$ N.

> $P = (T1 - T2)\ v = (1200 - 596) \times 15 = 9060$ W = **9.06 kW**.

III. Design of flat belts and pulleys

Belt materials

Material	Properties / use
Leather	Oak-tanned; high friction (μ 0.3-0.4), long life, costly; $\sigma \approx 2.5$ N/mm², density ≈ 950 kg/m³.
Rubberised fabric / cotton	Cheaper, good in damp; $\sigma \approx 1.5$-2.5 N/mm²; $\mu \approx 0.3$.
Balata / synthetic (nylon core)	Strong, thin, high speed, weather proof; modern flat belts with polyamide core and chrome leather.

Design procedure (power, speeds, centre distance given)

1. Design power = P × service factor Ks (1.1 to 1.8 for light to heavy shock, long hours).
2. Select belt speed v (pulley dia from speeds); find θ on smaller pulley and μ (corrected for speed if needed).

3. Find T1 and T2 from $P = (T1 - T2)\,v$ and $T1/T2 = e^{(\mu\theta)}$ (include Tc).
4. Allowable tension per unit width: choose thickness t (ply) and σ -> width $b = T1/(\sigma\, t\, \eta_{joint})$. Round to standard width.
5. Check pulley dia ≥ minimum ($d_{min} \approx 30$-35 t for leather, 50 t for fabric) so bending stress $\sigma b = E\, t/d$ is small.
6. Length of belt by formula; initial tension; check bearing loads = T1 + T2 (vector sum).

Flat pulley design

Part	Design rule
Material	Cast iron (most), steel, wood, aluminium for light. Speed < 30 m/s (CI rim).
Rim width B	$B = 1.1\,b + 10$ mm (about 1.25 b); thickness $t_r = d/200 + 3$ mm (single belt).
Crown	Slightly convex face (crown $h \approx B/100 - B/50$) so belt remains centred.
Arms	Elliptical section, $a2 = 0.5\,a1$ (minor/major); no. of arms n = 4 (d ≤ 200 mm), 6 (d 200-1500 mm). Arms designed as cantilever beams.
Arm design	Torque T is shared by n/2 arms acting as cantilevers of length $R = d/2$: $M = T / (n/2)$ ... $M = (T1 - T2)(d/2) / (n/2)$. Section modulus $Z = \pi\, a1^2\, a2/32 = M/\sigma b$ with $a2 = 0.5\,a1$ and $\sigma b = 15$ MPa (CI); so $a1 = [\,32\,M / (\pi\,(0.5)\,\sigma b)\,]^{(1/3)}$.
Hub	Bore = shaft dia ds; hub outer dia ≈ 2 ds; hub length ≈ (1.0 to 1.5) ds (or 0.5-0.75 B); standard key as per shaft size.
Balancing	Statically / dynamically balanced for v > 10 m/s.

> Rim stress due to rotation: $\sigma = \rho\, v^2$ (same as hoop stress) < 30 MPa (cast iron)

IV. V-belts and V-pulleys

V-belts are endless, trapezoidal belts made of rubber with fabric / polyester cords and wrapped cover. Used in **multiple belts** on one drive. Standard sections per IS 2494 / SAE: **Z, A, B, C, D, E**; automotive uses **AV 10, AV 13** narrow V and **serpentine (poly-V)** belts.

Section	Top width (mm)	Thickness (mm)	Min pulley dia (mm)	Power range (kW)
Z	10	6	50-70	0.2 - 1.5
A	13	8	75-125	0.7 - 3.5
B	17	11	125-200	2 - 15

Section	Top width (mm)	Thickness (mm)	Min pulley dia (mm)	Power range (kW)
C	22	14	200-300	7 - 70
D	32	19	355-500	20 - 150
E	38	23	500+	30 - 200

> Included angle of belt = 40°, groove angle = 34°-38° (smaller for small pulleys) so belt wedges and does not bottom.

> Advantages over flat belt: compact, high velocity ratio (≤ 7), less slip, short centre distance, drive in any direction, quiet.

> Limitations: not for very long distance, cannot use crossed drive, belts must be matched sets, more bending loss.

Design steps for V-belt drive (rated-power method)

1. Design power = P × Ks (service factor).
2. Select the **section** from the chart (design power vs speed of the faster shaft).
3. Choose the pitch dia of small pulley ≥ d_min; d2 = i d1 (1 - s). Belt speed v = π d1 N1/60 ≤ 25 m/s.
4. Centre distance: 0.7 (d1 + d2) ≤ C ≤ 2 (d1 + d2); compute pitch length L, choose the nearest standard length, then correct C.
5. Angle of lap θ = 180° - 57.3 (d2 - d1)/C -> correction factor Fθ (0.85 for 150°, 1.0 for 180°).
6. Rated power per belt (table) × Fθ × FL (length factor) = corrected power per belt.
7. **Number of belts** N = design power / corrected power per belt -> round up (max 8-12 belts).
8. Calculate pulley groove dimensions and check shaft loads and initial tension.

Example 2 - V-belt, number of belts

> P = 7.5 kW, motor 1440 rpm, driven 480 rpm, service factor 1.3. Design power = 9.75 kW. Section B selected; d1 = 140 mm -> d2 = 140 × 3 = 420 mm (no slip).

> v = π × 0.14 × 1440/60 = 10.5 m/s. Assume rated power per belt at this speed = 2.6 kW; Fθ = 0.94, FL = 1.0 -> corrected 2.44 kW.

> N = 9.75/2.44 = 4.0 -> **4 belts, section B.**

V-pulley details

> Cast iron or pressed steel; groove angle 34°/36°/38° chosen with pulley dia; face width = (n - 1) e + 2f for n belts (e = groove pitch, f = edge distance).

> Pulleys must be **aligned**, belts of same make/set, tension checked after 24 hrs of running.

V. Design of chain drives

A **roller chain** has inner plates, outer plates, pins, bushes and **rollers**; it wraps on sprockets (toothed wheels). Features: positive drive (no slip), efficiency 96-98%, high load capacity, long life, works in oil/dust; noisy at high speed, needs lubrication.

ISO chain no.	Pitch p (mm)	Roller dia (mm)	Min. breaking load (kN)
06B	9.525	6.35	8.9
08B	12.7	8.51	17.8
10B	15.875	10.16	22.2
12B	19.05	12.07	28.9
16B	25.4	15.88	60
20B	31.75	19.05	95
24B	38.1	25.4	160

Basic relations

Quantity	Formula
Pitch circle diameter	$D = p / \sin(180°/z)$ (z = number of teeth)
Velocity ratio	$i = N1/N2 = z2/z1$ ($z1 \geq 17$, preferably 19-25 odd; $z2 \leq 120$; $i \leq 7$)
Chain speed	$v = z1\ p\ N1 / 60000$ (m/s); v max about 15 m/s for roller chain
Chain length in pitches	$Lp = 2\ a/p + (z1 + z2)/2 + ((z2 - z1)/2\pi)^2\ (p/a)$ (round to even number)
Centre distance	$a = (30 \text{ to } 50)\ p$, max 80 p; $a_{min} = (D1 + D2)/2 + (30\text{-}50\ mm)$
Power	$P = (T1)\ v$ where tension in chain $F = P/v$; $P_{rated} \times K$ = design power
Load / safety factor	$n = Q / (Ks\ F_t + F_c + F_s) \geq 7\text{-}20$ depending on speed

Design procedure (selection method)

1. Assume z1 (19 to 27), $z2 = i\ z1$ -> find actual ratio.
2. Design power = $P \times Ks \times K_z / K_{multi}$ (Ks 1.0-1.5 service factor; multi-strand factor 1, 1.7, 2.5 for simplex, duplex, triplex).

3. Choose chain no. from rating chart (power vs sprocket rpm) and find pitch p.
4. Choose centre distance a = 30-50 p; calculate Lp (even number of links), correct a.
5. Check chain speed, tension, breaking-load factor of safety, sprocket tooth bearing stress.
6. Select lubrication method: (i) manual drip - v < 4 m/s, (ii) oil bath/splash v 4-10 m/s, (iii) forced oil jet v > 10 m/s.

> **Chordal action (polygon effect)**: chain moves like on a polygon, so speed varies slightly; reduces with larger teeth numbers; causes noise & impact -> keep z1 ≥ 17.

> Slack side tensioner / idler sprocket; chain guard; check wear elongation: replace at 3% extension.

VI. Wire ropes

A **wire rope** is made of many steel wires twisted into **strands** that are laid around a **core**. Designation **6 × 19**: 6 strands, each with 19 wires (flexible, used for cranes, lifts); 6 × 7 stiff (haulage), 6 × 37 very flexible.

Feature	Details
Wire material	Plough steel (1770-1960 MPa), improved plough steel, galvanised for corrosion.
Core	Fibre core (hemp/ sisal, flexible, holds lubricant) or **IWRC** (independent wire rope core, stronger, for heat/crushing).
Lay	**Ordinary (regular) lay**: wires opposite to strand lay; no kinking, general. **Lang's lay**: same direction; better wear but can untwist.
Rope geometry (6 × 19)	Rope dia d; wire dia dw = 0.063 d; metallic area A = 0.4 d^2 (mm^2); weight w ≈ 0.035 d^2 N/m; Ult. breaking load Fu ≈ 0.5 d^2 kN (typical).

Stresses in a wire rope

Stress	Formula
Direct (tension)	$\sigma d = (W + w\,L + Wa) / A = W_{total} / A$ (W = load, wL = rope weight, Wa = acceleration load)
Acceleration load	$Wa = (W + w\,L)\,a / g$
Bending stress	$\sigma b = E_r \cdot dw / D$ ($E_r = 0.8 \times 2.1 \times 10^5$ MPa; D = sheave / drum dia) -> larger D, lower stress
Total (max)	$\sigma_{max} = \sigma d + \sigma b$; **FOS** = Fu / ($W_{total}$ + bending-equivalent load)
Equivalent bending load	$Wb = \sigma b\,A$

Stress	Formula
Sheave / drum diameter	D/d ≥ 18 (absolute minimum), 30-45 for cranes with 6 × 19 rope; D ≥ 400 dw for 6 × 19 (check by bending stress)

> Factor of safety (typical): 5-6 cranes, 8-12 passenger lifts, 6-8 mine hoists, 3.5-5 for slings.

> **Selection steps**: find total load -> assume FOS -> pick rope dia d from Fu ≥ FOS × W_total -> choose D/d -> check bending, wear, fatigue (number of bends) and bearing pressure on sheave p = 2 W_t/(d D) ≤ allowable (5-9 MPa cast iron).

> Care: lubricate, avoid kinks, discard when 10% of wires broken in one lay length or 10% diameter loss.

VII. Lubrication and bearing design – basics

Bearing supports a rotating shaft and carries load with little friction. **Sliding (journal / plain) bearings** work on a fluid film; **rolling (ball/roller) bearings** on rolling contact. Automobile crank, cam and connecting-rod bearings are **hydrodynamic thin-shell sliding bearings**.

Lubrication regime	Film	Remark
Hydrodynamic (thick film)	Complete fluid film generated by motion (wedge)	Friction f = 0.001-0.005; needs speed; examples: crank journals
Hydrostatic	Film by external pump pressure	Works at zero speed, high load, machine tool spindles
Boundary	Surface asperities touch, thin adsorbed layer	Starting/stopping, f = 0.05-0.2
Mixed / thin film	Part fluid, part boundary	Low speed, high load
Elastohydrodynamic (EHL)	Elastic deformation + film in rolling/gears	Rolling bearings, gears, cams

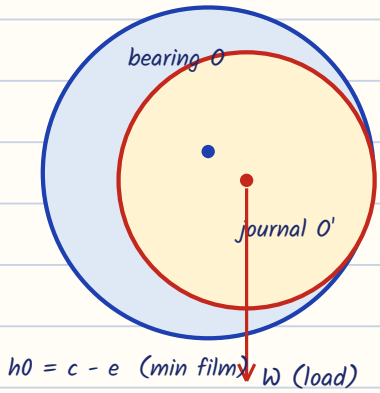
Viscosity

> Newton's law: shear stress $\tau = \mu (du/dy)$. **Absolute viscosity** μ (Pa·s = N s/m²); 1 Pa·s = 1000 cP = 10 poise; kinematic $\nu = \mu/\rho$ (mm²/s = cSt).

> Viscosity falls rapidly with temperature; **viscosity index (VI)** measures this change. SAE grades: SAE 5W-30, 10W-40, 15W-40 (engine); SAE 80W-90 (gear oil).

> **Petroff's equation** (lightly loaded, concentric journal): friction torque $T = 4\pi^2 \mu N r^3 l / c$; coefficient of friction $f = 2\pi^2 (\mu N/p)(r/c)$.

VIII. Hydrodynamic lubrication theory



c = radial clearance = R - r

e = eccentricity

ε = e / c (eccentricity ratio)

h0 = c (1 - ε)

At rest the journal touches the bearing; with speed, oil is dragged into the wedge and it **floats**.

Reynolds (1886) showed that a rotating journal drags oil into the converging wedge, creating a pressure that lifts it. **Osborne Reynolds' equation** for a one-dimensional (infinitely long) bearing is:

$$d/dx\ (\ h^3 / \mu \cdot dp/dx\ ) = 6\ U \cdot dh/dx$$

(U = surface velocity, h = film thickness)

> **Assumptions**: Newtonian incompressible lubricant, laminar flow, constant viscosity, no slip at the surfaces, pressure constant across film, negligible inertia & curvature, no side leakage (for 1-D).

> Pressure is max just after minimum film; full 3-D theory (finite bearing, l/d = 1) solved numerically by **Raimondi & Boyd charts** (variables: ε, h0/c, f r/c, Q/(r c N l), Ps etc. versus S).

> **Minimum film thickness** must exceed the sum of surface roughness: h0 ≥ 0.002 mm (≈ 5-10 times roughness) for safety.

> **Sommerfeld** solved Reynolds' equation exactly for the 'infinite bearing' (1904).

Design parameters (dimensionless)

Quantity	Formula / typical value
Unit bearing pressure	$p = W / (l\ d)$ (N/mm²); for automobile crank 7-14 MPa (up to 25 in main bearing)
Clearance ratio	$c / r = 0.001$ (to 0.002); $c = (d_{bearing} - d_{journal})/2$
Slenderness ratio	l / d = 0.5 to 2 (common 1 for crank: 0.5-0.75, machine 1-2)
Sommerfeld number	$S = (\mu N / p)(r / c)^2$ (N in rev/s, μ in Pa·s, p in Pa) - dimensionless
Eccentricity ratio	ε = e / c = 1 - h0 / c
Friction variable	f r/c = φ (S, l/d) from chart (Raimondi & Boyd)

Quantity	Formula / typical value
Heat generated	$Hg = f W v = f W (\pi d N)$ (W); $v = \pi d N$ (m/s, N rev/s)
Heat dissipated	$Hd = C A (t_b - t_a)$; $A = l d$ (projected area); C = 140-420 W/m²°C (still air), 490-1400 (good air flow)

Interpretation of Sommerfeld number

- **High S** (high speed, low load, high viscosity): thick film, small eccentricity ratio; **low S** (low speed, heavy load): film thin, $\varepsilon \to 1$, metal contact possible.
- Practical designs use S about 0.04 to 0.15 (stable side of the minimum-friction point); S below this is unsafe.
- Raimondi-Boyd chart gives h0/c, f r/c, flow Q, max pressure ratio for each S and l/d.

IX. Design of a sliding contact bearing using Sommerfeld number

1. Given: load W, journal dia d (from shaft design), speed N, operating temperature & lubricant (SAE grade) -> viscosity μ at operating temp.
2. Choose l/d (1 to 1.5), bearing material, allowable pressure p; $l = (l/d)\, d$; $p = W/(ld)$ must be ≤ allowable (check Table).
3. Choose clearance ratio c/r = 0.001 (or c = 0.001 d)(typical 0.0005 - 0.002).
4. Compute Sommerfeld number $S = (\mu N/p)(r/c)^2$; read from chart: h0/c, f r/c and flow for l/d.
5. Find $h0 = (h0/c)\, c$ and verify $h0 \geq (h0)min = 0.005 + 0.00004\, d$ (mm) ... i.e. ≥ about 0.02 mm.
6. Friction $f = (f\, r/c) / (r/c)$; heat generated $Hg = f W v$.
7. Heat dissipated $Hd = C A (tb - ta)$ with $A = l d$; check Hd ≥ Hg; if not, add cooling or oil circulation (or change l/d, c).
8. Final check: temperature rise of oil $\Delta t = Hg / (\rho Cp Q)$ with $\rho Cp \approx 1.76 \times 10^6$ J/m³K.

Bearing material	Max pressure (N/mm²)	Max temp (°C)	Use
Tin / lead babbitt	6 - 10	130	Crank, main bearings (thin lining on steel)
Copper-lead	14 - 20	150	Heavy diesel engines
Phosphor bronze	14	200	Gudgeon pins, gear boxes

Bearing material	Max pressure (N/mm²)	Max temp (°C)	Use
Aluminium-tin	28	150	Modern engine shells
Cast iron / carbon-graphite	3.5 – 7	up to 400	Slow speed, dry

X. Design using McKee's equation

McKee & McKee (1932) experimentally found that the coefficient of friction in a journal bearing varies with the **bearing characteristic number Z N / p** as shown by a curve with a **minimum friction point** – the point of transition between thick film and mixed lubrication.

$$\mu = K\,(Z\,N / p)(d / c) + k \quad \text{(McKee's equation)}$$

- Z = absolute viscosity (kg/m·s = Pa·s), N = speed (rpm), p = bearing pressure (N/mm²), d = journal dia, c = diametral clearance (same unit as d).
- K = 33 × 10^{-8} (constant) ; k = factor for end leakage: k = 0.002 for l/d from 0.75 to 2.8 (k depends on l/d; 0 for very long bearings).
- **Bearing modulus** K_b = value of (Z N/p) at the **minimum friction point**. For safe design operate at Z N/p ≥ 3 K_b – i.e. well in the thick film region – so that small changes of load or temperature do not cause film failure (metal contact).
- Heat generated Hg = μ W v (W); dissipated Hd = C A (t_b – t_a). With McKee, equate Hg = Hd to find operating temperature; check.
- K is read from the measured μ vs ZN/p curve (data-book); the margin of 3 ensures the bearing stays on the safe side of the minimum-friction point.

Example 3 – McKee check

> Journal d = 75 mm, l = 75 mm, W = 12 kN, N = 1500 rpm, Z = 0.02 kg/m·s (20 cP), diametral clearance c = 0.075 mm (d/c = 1000). $p = W/(l\,d)$ = 12000/(75 × 75) = 2.13 N/mm².

> $Z\,N / p$ = 0.02 × 1500 / 2.13 = 14.1. $\mu = 33 \times 10^{-8} \times 14.1 \times 1000 + 0.002 = 0.0047 + 0.002 =$ **0.0067**.

> $v = \pi\, d\, N/60$ = 5.89 m/s ; $Hg = \mu\, W\, v$ = 0.0067 × 12000 × 5.89 = **474 W**. $A = l\, d$ = 5.625 × 10^{-3} m²; with C = 1400 W/m²°C (ventilated): $Hd = 7.9\,(tb - ta)$ -> $tb - ta$ = 60 °C. Acceptable only with good oil circulation.

XI. Selection of rolling contact bearings

Type	Load capacity	Features / use
Deep-groove ball	Radial + moderate axial both ways	Most common; high speed, quiet; gearbox, wheels, motors
Angular contact ball	Radial + heavy axial (one way)	Used in pairs; machine tool spindles, pinion shafts
Self-aligning ball / spherical roller	Radial; accepts misalignment	Long shafts, heavy loads
Cylindrical roller	High radial, no/limited axial	Gearbox, rolls, heavy radial
Tapered roller	Radial + axial (one way)	Automobile wheel hubs, differential, pinion; mounted in pairs
Needle roller	High radial in small space	Universal joints, rockers, gearbox
Thrust ball / roller	Pure axial	Clutch release, vertical shafts

Life and capacity

Term	Meaning / formula
Basic rating life L10	Life (million revs) that 90% of identical bearings will reach or exceed (10% failure)
Basic dynamic load rating C	Constant radial load for which L10 = 1 million revolutions (from catalogue)
Life equation	L10 = (C/P)^k million revs; k = 3 (ball), 10/3 (roller)

Term	Meaning / formula
Life in hours	$L10h = 10^6 \, L10 / (60 \, n) = (10^6 / 60 \, n)(C/P)^k$
Equivalent dynamic load	$P = X \, V \, Fr + Y \, Fa$ (V = 1 inner ring rotating, 1.2 outer ring rotating)
Shock factor	$P' = Ks \, P$; Ks = 1.0 (steady), 1.2-1.5 (light shock), 1.5-3 (heavy shock)
Reliability factor	$L_R = a1 \, L10$; a1 = 1 (90%), 0.62 (95%), 0.21 (99%) (adjusted life)
Static load C0	Max load for which permanent deformation < 0.0001 d_roller; used for slow / oscillating
Varying loads	Equivalent load $P = [\, \Sigma (P_i^k \, n_i \, t_i) / \Sigma (n_i \, t_i) \,]^{(1/k)}$

Application	Required life L10h (hrs)
Automobile wheel / gearbox bearings	1500 - 5000 (about 3×10^5 km)
Light machinery, household	4000 - 8000
General industrial 8 hr/day	12,000 - 20,000
Continuous 24 hr (pumps, fans)	40,000 - 60,000

Selection procedure

1. Find radial & axial loads at the bearing (from gear forces), speed n, bore (shaft dia d).
2. Choose type (Table above) and required life L10h (application).
3. Compute equivalent load $P = X \, V \, Fr + Y \, Fa$ (X, Y from catalogue vs Fa/Fr and e).
4. Required dynamic capacity $C = P \, (60 \, n \, L10h / 10^6)^{(1/k)}$ (k = 3 ball, 10/3 roller).
5. Select a bearing (series, bore) from catalogue with $C_{catalogue} \geq C$ required; check static capacity and limiting speed.
6. Select fits (inner ring on shaft: k5/m5; outer in housing: H7), lubrication (grease/oil), sealing, preload, mounting arrangement (one fixed, one floating).

Example 4

> Deep-groove ball bearing, Fr = 4 kN, Fa = 0 -> P = 4 kN; n = 1200 rpm; L10h = 8000 hr.

> $L10 = 60 \times 1200 \times 8000/10^6 = 576$ million rev ; $C = P \times (L10)^{(1/3)} = 4 \times (576)^{(1/3)} = 4 \times 8.32 =$ **33.3 kN**.

> From the catalogue pick a deep-groove bearing with $C \geq 33.3$ kN, e.g. **6210 (C ≈ 35 kN)** (check actual catalogue value).

XII. Quick revision & important questions

Formulas to remember

> T1/T2 = e^(μθ) (flat); e^(μθ/sinβ) (V); P = (T1 - T2) v; Tc = m v²; max power at v = √(T_max/3m).

> Chain: D = p/sin(180°/z); v = z p N/60000; Lp = 2a/p + (z1 + z2)/2 + ((z2 - z1)/2π)² p/a.

> Rope: σd = W_total/A; σb = E_r dw/D.

> Sommerfeld S = (μN/p)(r/c)²; McKee μ = 33 × 10⁻⁸ (ZN/p)(d/c) + k; design at ZN/p ≥ 3K.

> Rolling bearing: L10 = (C/P)^k; P = X V Fr + Y Fa.

1. Derive the ratio of belt tensions for a flat belt and a V-belt. (5 M)
2. Design a flat belt drive to transmit given power; also design the pulley. (10 M)
3. Explain the design of a V-belt drive. Selection of number of belts. (10 M)
4. Design a roller chain drive for given power and speeds. (10 M)
5. A wire rope is used in a hoist; find rope size and stresses. (10 M)
6. Explain hydrodynamic lubrication theory with Reynolds equation. (10 M)
7. Design a journal bearing using Sommerfeld number / McKee's equation. (10 M)
8. How do you select a rolling contact bearing? Explain L10 life. (10 M)

UNIT II

Spur Gear Design

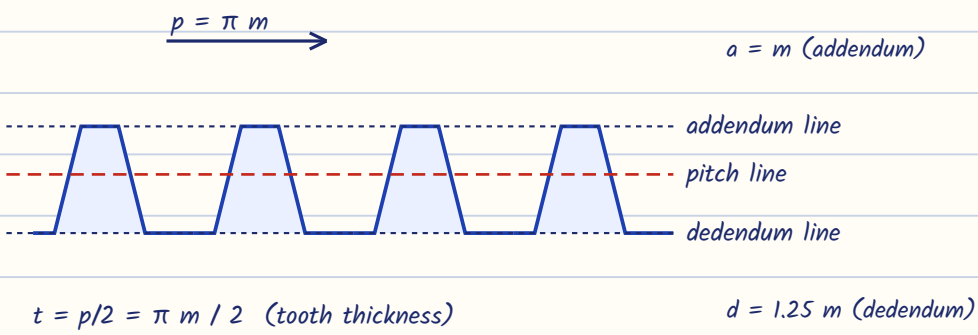
gear geometry | kinematics | forces on tooth | stresses | material selection (bending & contact) | design | power capacity

I. Introduction to gears

Gears are toothed wheels that transmit power and motion between shafts with a **constant velocity ratio** and no slip. In an automobile, gears are used in the **gearbox, differential, final drive, timing drive and steering**. **Spur gears** have teeth cut **parallel to the shaft axis**; they connect **parallel shafts**; simplest and cheapest, but noisy at high speed.

Classification	Types
By shaft position	Parallel: spur, helical, herringbone, rack & pinion. Intersecting: bevel. Non-parallel non-intersecting: worm, crossed helical, hypoid.
By peripheral velocity	Low (< 3 m/s), medium (3-15 m/s), high (> 15 m/s).
By meshing	External, internal (annular), rack and pinion.
By tooth profile	Involute (standard), cycloidal.

II. Gear geometry



Term	Meaning / relation
Pitch circle (d)	Imaginary circle that rolls without slip with the mating circle. $d = m\,z$
Module (m)	$m = d / z$ (mm); the basic size of the tooth. Standard (IS 2535): 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20 …
Circular pitch (p)	Arc distance between corresponding points of adjacent teeth on pitch circle: $p = \pi d / z = \pi m$
Diametral pitch (P)	$P = z / d = 1 / m$ (teeth per inch in inch system)
Addendum (ha) / **Dedendum** (hf)	Radial heights above / below pitch circle: $ha = m$; $hf = 1.25\,m$ (for 20° full depth)
Clearance (c)	$c = hf - ha = 0.25\,m$
Tooth thickness / space	$t = p/2 = \pi m / 2$ (theoretical); backlash = space width - mating tooth thickness
Pressure angle (φ)	Angle between the common normal (line of action) and the common tangent at the pitch point; standard 20° (14.5° older)
Base circle (db)	$db = d \cos \varphi$: circle from which the involute is generated; base pitch $pb = \pi m \cos \varphi$
Outside / root diameter	$da = m (z + 2)$; $df = m (z - 2.5)$
Centre distance (a)	$a = (d1 + d2)/2 = m (z1 + z2)/2$
Face width (b)	Width of tooth along the axis: $b = 9.5\,m$ to $12.5\,m$ ($\approx 10\,m$ typical); $b \le d1$ (pinion)
Gear ratio (i)	$i = z2/z1 = N1/N2 = d2/d1 \ge 1$ (reduction); up to 6-8 per stage for spur gears

Standard tooth systems

System	φ	Addendum	Dedendum	Remark
14.5° composite / full depth	14.5°	m	1.25 m	Old; min 32 teeth to avoid interference
20° full depth	20°	m	1.25 m	Most used; stronger tooth, min 17 teeth
20° stub	20°	0.8 m	1.0 m	Very strong, short; min 14 teeth; less sliding

III. Law of gearing and involute profile

> **Law of gearing**: for a constant velocity ratio, the **common normal at the point of contact must always pass through the fixed pitch point** P on the line of centres.

> **Involute** = locus of a point on a taut string unwrapped from a base circle. It satisfies the law; the line of action is the straight line tangent to both base circles (inclined at φ).

> Advantages: **centre distance can vary slightly without affecting velocity ratio**, easy to manufacture by hobbing, constant pressure angle.

> Involute function: inv φ = tan φ − φ (radians). Involute pressure angle at radius r: cos φ_r = rb/r.

Kinematics of meshing

Quantity	Formula
Velocity ratio	N1/N2 = ω1/ω2 = d2/d1 = z2/z1
Pitch line velocity	v = π d1 N1 / 60000 (m/s, d in mm, N in rpm)
Path of contact	Approach + recess: = √(Ra1² − Rb1²) + √(Ra2² − Rb2²) − a sin φ
Contact ratio (CR)	CR = (length of path of contact) / (base pitch pb) ; for spur gears ≥ 1.2 (1.4–1.6 good). It = average number of teeth in contact.
Sliding velocity	Zero at pitch point; max at start & end of contact; Vs = (ω1 + ω2) × (distance from pitch point)

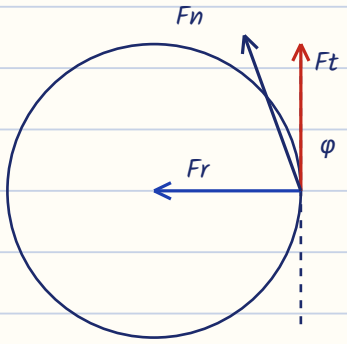
Interference & undercutting

> **Interference**: the tip of one tooth cuts the root of the other because contact occurs beyond the tangent points (base circles). It causes undercutting -> weakens the tooth.

> Minimum teeth to avoid interference on a pinion meshing with a rack: z_min = 2 ha*/sin²φ = **17 (20° FD), 14 (20° stub), 32 (14.5°)**. Smaller number is possible against a larger gear (about 13 for equal 20° gears).

> Remedies: stub teeth, larger pressure angle, **profile shift (addendum modification)**, undercut the pinion root on purpose, increase the pinion teeth.

IV. Forces on a gear tooth



Ft = 2T / d = 2000 P / (π d N/60)

Fr = Ft tan φ

Fn = Ft / cos φ

Ft transmits power; Fr only loads shaft & bearings (no work done).

φ = pressure angle (20°)

Force	Formula (spur gear)
Torque on pinion	$T = 60\,P / (2\pi N) = 9.55 \times 10^6\,P / N$ (N·mm; P in kW, N in rpm)
Tangential force Ft	$Ft = 2\,T / dl = 2\,T / (m\,zl) = 1000\,P / v$ (v in m/s)
Radial force Fr	$Fr = Ft \tan \varphi$
Normal (total) force Fn	$Fn = Ft / \cos \varphi = \sqrt{(Ft^2 + Fr^2)}$
Load on shaft/bearing	Resultant Fn; shaft is bent by Fr (in plane) and Ft (perpendicular plane) and twisted by T.

> On the driven gear the forces are equal and opposite; if the pinion is overhung, the shaft bending increases.

> Using direction: driving gear force opposes rotation at tooth; shaft reactions by taking moments in the plane of Ft and in the plane of Fr.

Example 1 – forces

> P = 7.5 kW at 1000 rpm; zl = 20, m = 3 mm, φ = 20°. $T = 9.55 \times 10^6 \times 7.5/1000 = 71{,}625$ N·mm.

> dl = 60 mm -> Ft = 2T/dl = 2387 N ; Fr = 2387 tan 20° = 869 N ; Fn = 2387/cos 20° = **2540 N**.

V. Stresses in gear tooth

1. Bending stress – Lewis equation

Wilfred Lewis (1892) treated the tooth as a **cantilever beam** loaded by Ft at the tip, with a parabola (weakest section) of thickness t at height h: $\sigma b = M/Z = (Ft\ h) / (b\ t^2/6) = 6\ Ft\ h / (b\ t^2)$. Writing $t^2 = 4 \times h$ ... gives:

$$\sigma b = Ft / (b\ m\ Y) \quad \text{with} \quad Y = \pi\ y \ \text{(Lewis form factor)}$$

z	15	17	20	25	30	40	60	100	rack
y (20° FD)	0.093	0.100	0.108	0.117	0.124	0.131	0.139	0.145	0.154

> Formulae: $y = 0.154 - 0.912/z$ (20° FD); $y = 0.124 - 0.684/z$ (14.5° composite); $y = 0.175 - 0.841/z$ (20° stub).

> **Beam strength** $Sb = \sigma b\ m\ b\ Y$ (N) – the load the tooth can carry without breaking. The weaker of pinion / gear (smaller $\sigma b \times Y$) governs: compare $\sigma b1\ y1$ and $\sigma b2\ y2$.

> Valid for the load at the tip – conservative; actual contact at pitch point of two teeth pairs (CR > 1).

2. Dynamic load (velocity factor)

Errors in profile and pitch, elasticity and inertia produce **impact**, so the actual load is higher than Ft.

Barth's equation uses a velocity factor Cv:

Gear quality	Cv (v in m/s)	Range
Ordinary / cut gears	$Cv = 3 / (3 + v)$	v < 10 m/s
Carefully cut, hobbed	$Cv = 4.5 / (4.5 + v)$	v up to 20 m/s
Accurately hobbed & generated	$Cv = 6 / (6 + v)$	v up to 20 m/s
Precision ground / shaved	$Cv = 5.6 / (5.6 + \sqrt{v})$	v > 20 m/s

> **Effective (dynamic) load** $Fd = Ft / Cv$ (or $Cs\ Ft / Cv$ with service factor Cs: 1.0 uniform, 1.25 light shock, 1.5 medium, 1.75–2.0 heavy).

> **Buckingham**: $Fd = Ft + Fi$; $Fi = 21\ v\ (C\ b + Ft) / (21\ v + \sqrt{(C\ b + Ft)})$, with C = deformation factor ≈ 11,860 e for steel pair (20° FD), e = tooth error (mm).

> Design rule: **Sb ≥ Fd** (bending) and **Sw ≥ Fd** (wear).

3. Surface (contact) stress – wear

Pitting (surface fatigue) is caused by Hertz contact stress between two cylinders of radii ρ1 and ρ2 at the pitch point ($\rho = (d/2) \sin \varphi$):

$$\sigma c = \sqrt{[\ (Fn / \pi\ b)\ (1/\rho 1 + 1/\rho 2)\ /\ (\ (1 - \nu 1^2)/E1 + (1 - \nu 2^2)/E2\ )\ ]} \leq \sigma c,allow$$

> For steel-steel ($E = 2.06 \times 10^5$ MPa, $\nu = 0.3$): $\sigma c \approx 190 \sqrt{[\ Fn\ (1/\rho 1 + 1/\rho 2)\ /\ b\ ]}$ (MPa; N, mm).

> **Buckingham wear load** $Sw = b\ Q\ d1\ K$ where $Q = 2\ z2/(z1 + z2) = 2\ i/(i + 1)$ (ratio factor; external gears) and $K = (\sigma es)^2 \sin\varphi\ (1/E1 + 1/E2)/1.4$ = load-stress factor.

> Approximately $K = 0.16\ (BHN/100)^2$ N/mm² for steel on steel (BHN = average hardness of the pair, or softer member).

> σes = surface endurance limit ≈ (2.8 BHN - 70) MPa for steel (take other materials from the data book).

> Condition: **Sw ≥ Fd**.

VI. Selection of gear material

Material must resist **bending fatigue at root** (strength, toughness) and **pitting wear at flank** (surface hardness), be machinable, damping, cheap. Choose according to **σb (bending stress)** and **σc / BHN (contact stress)**.

Material	σut (MPa)	BHN	Allowable σb ≈ σut/3 (MPa)	Use
Grey cast iron FG200	200	180	67	Low-cost, slow, quiet; large gears
Grey cast iron FG260	260	200	87	Medium duty, good damping
Steel C30 / C40 (normalised)	500-580	150-180	170-190	General, light-medium loads
Steel C45 / 40C8 (hardened & tempered)	600-750	200-250	200-250	Medium - heavy; most common
Alloy steel 40Cr1 / 40Ni3	800-1000	250-300	270-330	Heavy duty, shock
Case-hardened 20MnCr5 / 16MnCr5 / 17Cr	900-1100 (core)	surface 55-62 HRC	(case) 300-400	**Automobile gears**, high power density
Nitrided steel (31CrMoV9)	1000+	surface 60 HRC	300-400	Minimum distortion, high precision
Phosphor bronze, Nylon, phenolic (textolite)	low	-	low	Worm wheels; quiet, lightly loaded; timing gears

> Values are typical; use the exact allowable stresses from the PSG / Bhandari design data book for the problem.

> **Pinion harder than gear** by 30-50 BHN (the pinion has more cycles). For same material -> pinion is weaker in bending only if σb1 y1 < σb2 y2: check **σ y** for both.

> **Selection logic**: (1) estimate Ft; (2) compare Sb and Sw requirement; (3) if wear governs raise hardness (heat treatment, case hardening); if bending governs raise σut or module/face width.

> Heat treatments: through-hardening (BHN ≤ 350), flame/induction hardening, carburising + quench (58-62 HRC), nitriding.

VII. Design of spur gears

Given: power P, pinion speed N1, gear ratio i, working conditions (shock, hours). **Required**: module m, teeth z1 z2, face width b, material, checks.

1. Choose **materials** and heat treatment -> σb1, σb2, BHN. Choose pressure angle 20° FD, $z1 \geq 17$ (usually 18-24), $z2 = i\ z1$ (round).
2. Design torque $T = 9.55 \times 10^6\ P\ Cs / N1$ (N·mm) with service factor Cs.
3. Assume face width $b = 10\ m$ ($b/m = 9.5$-12.5). Pitch line velocity $v = \pi\ m\ z1\ N1/60000$ -> Cv.
4. Find y1, y2 -> $Y = \pi\ y$; weaker gear = smaller (σb × y). Design the **weaker** one for **beam strength**: $Ft = \sigma b\ Cv\ b\ \pi\ y\ m$.
5. Solve for module: from $Ft = 2T/(m\ z1)$ and $Ft = \sigma b\ Cv\ (b/m)\ \pi\ y\ m^2$:

$$m^3 = 2\ T / [\ z1\ \sigma b\ Cv\ (b/m)\ \pi\ y\ ] \quad \text{-> take next standard module}$$

6. Fix $b = 10\ m$, $d1 = m\ z1$, $d2 = m\ z2$, $a = (d1 + d2)/2$, recompute v, Cv, $Fd = Ft/Cv$.
7. **Check wear**: $Sw = b\ Q\ d1\ K \geq Fd$; if not, increase hardness / b / m or change material.
8. Check bending: $Sb = \sigma b\ m\ b\ Y \geq Fd$. Also check **contact ratio, gear blank dimensions, shaft & key**.
9. Specify: module, z, face width, material & heat treatment, grade of accuracy, lubrication.

Example 2 – complete design

> Design a pair of spur gears to transmit 7.5 kW at 1000 rpm, $i = 3$, 20° full depth. σb (both) = 160 MPa; $z1 = 20$, $z2 = 60$, $Cs = 1$, ordinary cut gears ($Cv = 3/(3 + v)$).

> $T = 9.55 \times 10^6 \times 7.5/1000 = 71{,}625$ N·mm ; $y = 0.154 - 0.912/20 = 0.1084$; $Y = \pi y = 0.3405$; $b = 10\ m$.

> $m^3\ Cv = 2T/[z1\ \sigma b\ (b/m)\ \pi\ y] = 143{,}250/[20 \times 160 \times 10 \times 0.1084 \times \pi] = 13.15$. Pitch velocity $v = 1.047\ m$ (m/s). Try $m = 3$: $v = 3.14$ m/s, $Cv = 3/6.14 = 0.489$, $m^3\ Cv = 27 \times 0.489 = 13.2$ ✓ -> **$m = 3$ mm**.

> $d1 = 60$, $d2 = 180$, $a = 120$ mm, $b = 30$ mm. $Ft = 2T/d1 = 2388$ N ; $Fd = Ft/Cv = 4886$ N. $Sb = 160 \times 3 \times 30 \times 0.3405 = 4903\ N \geq Fd$ ✓.

> Wear: $Q = 2 \times 3/4 = 1.5$; for BHN = 350, $K = 0.16\ (3.5)^2 = 1.96$; $Sw = b\ Q\ d1\ K = 30 \times 1.5 \times 60 \times 1.96 = 5292\ N \geq 4886$ ✓ -> pinion & gear hardness about 350 BHN.

VIII. Power transmitting capacity

Power capacity of a given pair = the maximum power it can transmit safely. It is limited by the **lower** of the beam strength and wear strength.

$$P = Ft_allow \cdot v / 1000 \ \ (kW); \quad Ft_allow = Cv \cdot \min(Sb, Sw) / Cs$$

> Use the **weaker member** (pinion or gear) for each failure mode: $Sb = \sigma b\ m\ b\ Y$; $Sw = b\ Q\ d1\ K$.

> Power $\propto v$, so power capacity rises with speed up to the point where dynamic load (Cv) cuts it: capacity is maximum near $v = 10$–15 m/s for ordinary gears.

> Increase capacity by: harder surface, larger module/face width ($b \leq 14\ m$ to avoid misalignment), better accuracy (higher Cv), better lubrication, helical gears.

Example 3 – power capacity

> Using the gear pair of Example 2: $Sb = 4903$ N, $Sw = 5292$ N -> min = 4903 N. $Ft_allow = Cv \times 4903 = 0.489 \times 4903 = 2398$ N.

> $P = Ft_allow \times v / 1000 = 2398 \times 3.14 / 1000 =$ **7.53 kW** ≈ design power ✓.

IX. Gear tooth failures

Failure	Cause	Remedy
Tooth breakage	Overload / bending fatigue at root; shock; misalignment	Larger module / face width, tougher steel, fillet radius, accurate mounting
Pitting	Contact (Hertz) fatigue of the surface near pitch line	Higher hardness, better lubricant, larger gear dia, improved finish
Scoring / scuffing	Breakdown of oil film at high speed & load; welding of asperities	EP additives, lower temp, surface treatment, smoother profile
Abrasive wear	Dirt / foreign particles in the lubricant	Filtration, seals, clean oil
Plastic flow / rolling	Overload on soft material (cast iron / soft steel) flows the surface	Harder material, reduce load
Corrosive wear	Acids / moisture	Correct lubricant, sealing, coatings

Lubrication and general design rules

> **Splash** lubrication for v < 12 m/s (oil level 1-3 teeth deep); **pressure spray / jet** for higher speeds. Oil: SAE 80W-90 / EP gear oil (gearbox).

> Gear wheel construction: solid disc (d < 200 mm), web with holes (200-500 mm), arms (> 500 mm, elliptical 4-6 arms, cast).

> **Backlash** (clearance between non-working flanks) ≈ 0.04 m - 0.1 m for smooth running and thermal expansion.

> Accuracy grade (IS 3756): grade 6-8 for automobile gears; hobbing, shaping, shaving, grinding to improve.

> Face width limit: b ≥ 9.5 m (good contact) and b ≤ 12.5 m typical. Pinion with **z1 ≥ 17** to avoid undercut.

IX-A. Additional topics & solved examples

Contact ratio - worked example

z1 = 20, z2 = 60, m = 3, φ = 20°. r1 = 30, r2 = 90, a = 120 mm. Ra1 = 33, Ra2 = 93 ; Rb1 = 30 cos 20° = 28.19, Rb2 = 84.57. Path of contact = $\sqrt{(33^2 - 28.19^2)} + \sqrt{(93^2 - 84.57^2)} - 120 \sin 20° = 17.15 + 38.69 - 41.04 = 14.80$ mm. Base pitch $p_b = \pi m \cos \varphi = 8.857$ mm. **CR = 14.80/8.857 = 1.67** (> 1.2 ✓ - 1 to 2 pairs in contact).

Which gear is weaker? - comparison of σ y

Pinion C45 (σb = 200 MPa) z1 = 18: y1 = 0.154 - 0.912/18 = 0.1033 -> σb1 y1 = 20.7. Gear grey CI FG260 (σb = 87 MPa) z2 = 54: y2 = 0.154 - 0.912/54 = 0.1371 -> σb2 y2 = 11.9. Smaller product = **gear is weaker**, so the module is designed for the cast-iron gear (use its σb and y in Ft = σb Cv b π y m); then check the pinion.

Profile shift (addendum modification)

- Move the cutter away from the blank by x m: **positive shift** thickens the root of a small pinion and avoids undercut.
- Minimum shift to avoid undercut: x_min = (17 - z)/17 (20° FD). Eg z = 12: x = 0.29 m.
- Centre distance changes with shifts; sum of shifts (x1 + x2) = 0 keeps a = standard (equal & opposite shifts: 'S0' gears).

Rack & pinion, internal gears

Gear type	Notes
Rack and pinion	Rack = gear of infinite radius; straight teeth. Linear travel = π d N; steering gear.
Internal gear	Teeth on inside of a ring; centre distance a = m (z2 - z1)/2; both rotate in same direction; compact, strong (Q = 2 z2/(z2 - z1)); used in planetary gear trains.
Idler gear	Does not change ratio, only direction: reverse gear in automobile gearbox.

Manufacture of gears

Method	Process	Quality
Casting	Sand casting, die casting, investment casting	Rough; large slow gears
Forming	Form milling cutter (one cutter per teeth no. range)	Moderate; job work
Generating	Hobbing, gear shaping, rack cutters - mass production, accurate involute	Good (most used)
Finishing	Shaving, grinding, lapping, honing, burnishing	High accuracy, quiet automotive gears
Others	Powder metallurgy (sintered), forging, plastic moulding (nylon), rolling	Light-duty gears, cheap

Example 4 - second design problem (with service factor)

> Design a spur gear pair for P = 10 kW, N1 = 720 rpm, i = 2.5, medium shock (Cs = 1.25), steel σb = 140 MPa, 20° FD, z1 = 20, z2 = 50, b = 10 m.

> Design torque $T = 9.55 \times 10^6 \times 10 \times 1.25/720 = 165{,}800$ N·mm. y = 0.1084. m^3 Cv = 2T/[z1 σb (b/m) π y] = 331,600/[20 × 140 × 10 × 0.1084 × π] = 34.8.

> m = 4 -> v = 3.02 m/s, Cv = 0.499, m^3 Cv = 31.9 < 34.8 ✗; m = 5 -> v = 3.77, Cv = 0.443, m^3 Cv = 55.4 ✓ -> **m = 5 mm**.

> d1 = 100, d2 = 250, a = 175 mm, b = 50 mm. Ft = 2T/d1 = 3316 N ; Fd = Ft/Cv = 7483 N ; Sb = 140 × 5 × 50 × 0.3405 = 11,918 N ✓.

> Wear: Q = 2 × 50/70 = 1.43 ; need K ≥ 7483/(b Q d1) = 7483/(50 × 1.43 × 100) = 1.05 -> 0.16 $(BHN/100)^2 \geq 1.05$ -> **BHN ≥ 256**. Choose C45 hardened & tempered, BHN 260 (Sw = 7726 N ✓).

Common mistakes to avoid in exam

> Using module in metres with forces in newtons - keep **mm and N** (stress in N/mm^2 = MPa).

> Forgetting the **service factor** when finding torque, or using Cv with the wrong velocity formula for the gear quality.

> Designing the stronger gear: always design the **weaker** (smaller σb × y).

> Not rounding module to the **standard value** and recomputing d, v, Cv and the checks.

> Checking only bending - wear (pitting) often controls for hardened pinions.

Efficiency, shaft design for gears

> Efficiency of a spur-gear pair (well lubricated) = **98-99%**; torque on gear T2 = i T1 η.

> Shaft: resultant bending moment from Ft (horizontal plane) & Fr (vertical plane): $M = \sqrt{(Mh^2 + Mv^2)}$; design with ASME code $d^3 = (16/\pi\ \tau)\ \sqrt{[(Kb\ M)^2 + (Kt\ T)^2]}$.

> Overhung pinions are avoided; keep bearing span so that deflection < 0.0005 rad at the gear for proper tooth contact.

> Gear blank: rim thickness ≥ 1.5-2 × tooth height; hub dia = 1.8 d_shaft; hub length = 1.2-1.5 d_shaft.

Modern (AGMA / ISO) approach in brief

> Bending stress: σ = Ft Ko Kv Ks Km / (b m J) - factors for overload, dynamics, size, load distribution; J = geometry factor.

> Contact stress: σc = Cp √(Ft Ko Kv Ks Km Cf / (b d1 I)) with Cp = 191 √MPa (steel pair); I = geometry factor for pitting.

> Allowable stresses corrected for life (cycles), reliability, temperature; safety factors 1.0–1.5 for bending and 1.0–1.2 for contact.

X. Quick revision & important questions

Formulas to remember

> $d = m\,z$; $p = \pi\,m$; $ha = m$; $hf = 1.25\,m$; $a = m\,(z1 + z2)/2$; $db = d\cos\varphi$; $z_{min} = 17$ ($20°$ FD).

> $T = 9.55 \times 10^6\,P/N$; $Ft = 2T/d$; $Fr = Ft\tan\varphi$; $Fn = Ft/\cos\varphi$; $v = \pi\,d\,N/60000$.

> $\sigma b = Ft/(b\,m\,Y)$, $Y = \pi\,y$; $y = 0.154 - 0.912/z$; $Sb = \sigma b\,m\,b\,Y$; $Fd = Ft/Cv$; $Cv = 3/(3 + v)$.

> $Sw = b\,Q\,d1\,K$; $Q = 2z2/(z1 + z2)$; $K = 0.16\,(BHN/100)^2$; Design: $Sb, Sw \geq Fd$.

> $m^3 = 2T/(z1\,\sigma b\,Cv\,(b/m)\,\pi\,y)$; $P = Ft\,v/1000$.

1. Explain gear terminology with a neat sketch and define module, pressure angle. (5 M)
2. State the law of gearing. What is interference? How can it be avoided? (5 M)
3. Derive Lewis equation for bending stress in spur gear tooth. (10 M)
4. Explain the dynamic load on gear tooth (Barth and Buckingham). (5 M)
5. Derive Buckingham wear load / Hertz contact stress for gears. (10 M)
6. How do you select a gear material based on bending and contact stresses? (5 M)
7. Design a spur gear pair for given power, speed and ratio. (10 M)
8. Determine the power transmitting capacity of a given spur gear pair. (10 M)

UNIT III

Helical, Bevel and Worm Gears

parallel helical | crossed helical | straight bevel | worm gearing | kinematics | forces | stresses | friction & efficiency | design

I. Parallel helical gears

Helical gears have teeth cut at a **helix angle β** to the axis. Contact starts at one end of the tooth and spreads gradually along an inclined line, so the load is picked up smoothly - **quieter, stronger, higher speed** (up to 50 m/s) than spur gears, but they produce **axial thrust**. Used in automobile **constant-mesh gearbox**, differential final drives, timing drives.

Feature	Spur	Helical
Contact	Line across whole face at once	Gradual along inclined line; 2-3 teeth in contact
Noise / smoothness	Noisy at high speed	Quiet, smooth
Load capacity	Lower	Higher (longer contact line, higher contact ratio)
Axial thrust	None	Yes: $F_a = F_t \tan \beta$ (use thrust/taper bearings, or double helical)
Min. teeth (20° FD)	17	$17 \cos^3\beta$ (about 15-16 at β = 15-20°)
Cost	Cheap	Dearer (cutting with helical hob / cutters)

Kinematics and tooth proportions

Quantity	Formula
Hand of helix	Parallel shafts: pinion RH ↔ gear LH (opposite hands, same angle β)
Normal module / circular pitch	m_n ; $p_n = \pi m_n$ (standard values, cutter cuts normal section)
Transverse module / pitch	$m_t = m_n / \cos\beta$; $p_t = \pi m_t = p_n / \cos\beta$
Axial pitch	$p_x = p_t / \tan\beta = \pi m_n / \sin\beta$
Pressure angles	$\tan\varphi_t = \tan\varphi_n / \cos\beta$ (φ_n = 20°)
Pitch diameter	$d = m_t z = m_n z / \cos\beta$
Centre distance	$a = m_n (z1 + z2) / (2 \cos\beta)$ (adjustable by changing β)

Quantity	Formula
Virtual (formative) teeth	$z' = z / \cos^3\beta$ - the equivalent spur gear in the normal plane (used in Lewis equation)
Velocity ratio, speed	$i = z2/z1 = N1/N2$; $v = \pi\, d1\, N1 / 60000$
Face width	$b \geq px = \pi\, mn/\sin\beta$ (overlap) for helical action; usually $b = 10\ mn$ to $12.5\ mn$ or $b \geq 1.15\ px - 2\ px$
Helix angle	$\beta = 15°\text{-}30°$ (single helical); 30°-45° for double helical (herringbone)
Contact ratio	Total CR = transverse CR + axial CR (b/px) ; total ≥ 2 is very smooth

Force analysis on a helical gear tooth

The normal force Fn lies in the normal plane; resolve it into three perpendicular components at the pitch point:

Component	Formula	Effect
Tangential Ft	$Ft = 2T / d1 = 2000\ P / v$	Transmits power; bends tooth, twists shaft
Radial Fr	$Fr = Ft \tan\varphi n / \cos\beta = Ft \tan\varphi t$	Separating force; bends shaft
Axial (thrust) Fa	$Fa = Ft \tan\beta$	Thrust bearing load; bends shaft with moment Fa d/2
Normal force Fn	$Fn = Ft / (\cos\varphi n \cos\beta)$	Resultant

> Larger β -> smoother but larger thrust. Thrust direction depends on hand of helix and rotation; choose hand so that thrust loads the shaft bearing with the larger capacity.

> **Double helical (herringbone)** gears: two opposite helices on one blank; axial thrusts **cancel** inside the gear, so high β (up to 45°) can be used - heavy-duty reducers, turbines.

Stresses in helical gear tooth

$$\sigma b = Ft / (b\ mn\ Y') \quad \text{with} \quad Y' = \pi\, y', \quad y' = 0.154 - 0.912 / z' \quad (z' = z / \cos^3\beta)$$

> **Beam strength** $Sb = \sigma b\ mn\ b\ Y'$; dynamic load $Fd = Cs\ Ft / Cv$ ($Cv = 3/(3 + v)$, $4.5/(4.5 + v)$ or $6/(6 + v)$ as for spur gears).

> **Wear (Buckingham)** $Sw = b\ Q\ d1\ K / \cos^2\beta$ with $Q = 2\ z2/(z1 + z2)$, $K = 0.16\ (BHN/100)^2$.

> Check **Sb ≥ Fd** and **Sw ≥ Fd**; the weaker of pinion/gear ($\sigma b\ y'$) governs.

> Helical gears carry about **30–50% more power** than equal spur gears because of the longer line of

contact.

Design of helical gears - steps

1. Select β (15°-25° typical), $\varphi n = 20°$, material (σb, BHN), $z1$ ($\geq 17 \cos^3\beta$), $z2 = i\ z1$.
2. $T = 9.55 \times 10^6\ P\ Cs / N1$; assume $b = 10\ mn$; $z1' = z1/\cos^3\beta$ -> $y1'$ (and $y2'$).
3. $d1 = mn\ z1 / \cos\beta$; $v = \pi\ d1\ N1 / 60000$ -> Cv. Design the weaker gear for strength:

$$mn^3 = 2\ T \cos\beta\ /\ [\ z1\ \sigma b\ Cv\ (b/mn)\ \pi\ y'\ ] \quad \rightarrow \quad \text{standard } mn$$

4. Calculate d1, d2, $a = (d1 + d2)/2$, b; check $b \geq px$ (axial overlap).
5. $Fd = Ft / Cv$; check $Sb \geq Fd$ and wear $Sw = b\ Q\ d1\ K / \cos^2\beta \geq Fd$ -> required BHN.
6. Find axial thrust Fa, radial Fr and select bearings; check shaft design.

Example 1 - helical gear design

> P = 10 kW, N1 = 1000 rpm, i = 3, $\beta = 20°$, $\varphi n = 20°$, z1 = 20, z2 = 60, σb = 160 MPa (both), ordinary cut gears. T = 95,500 N·mm.

> $z1' = 20/\cos^3 20° = 24.1$ -> $y' = 0.154 - 0.912/24.1 = 0.1162$; $Y' = \pi\ y' = 0.365$. $d1 = mn \times 20/0.9397 = 21.28\ mn$; $v = 1.114\ mn$ (m/s).

> $mn^3\ Cv = 2T \cos\beta/[z1\ \sigma b\ (b/mn)\ \pi\ y'] = 2 \times 95{,}500 \times 0.9397/(20 \times 160 \times 10 \times 0.365) = 15.4$. mn = 3: v = 3.34, Cv = 0.473, $mn^3\ Cv = 12.8$ ✗; mn = 4: v = 4.46, Cv = 0.402, $mn^3\ Cv = 25.7$ ✓ -> **mn = 4 mm**.

> d1 = 85.1, d2 = 255.4, a = 170.3 mm; $b = 40\ mm \geq px = \pi \times 4/\sin 20° = 36.7$ ✓. $Ft = 2T/d1 = 2244$ N; $Fd = Ft/Cv = 5582$ N; $Sb = 160 \times 4 \times 40 \times 0.365 = 9344$ N ✓.

> Wear: Q = 1.5; need $K \geq Fd \cos^2\beta/(b\ Q\ d1) = 5582 \times 0.883/(40 \times 1.5 \times 85.1) = 0.965$ -> BHN ≥ 246 -> **BHN 250**. $Fa = Ft \tan 20° = 817$ N; $Fr = Ft \tan 20°/\cos 20° = 869$ N.

II. Crossed helical gears (screw gears)

Two helical gears mounted on **non-parallel, non-intersecting shafts**. They have **point contact** (the teeth are tangent at a point), so the load capacity is **low** and sliding is high. Used for **light drives** - oil pumps, distributor / speedometer drives in automobiles, instruments.

Item	Relation
Shaft angle	$\Sigma = \beta1 + \beta2$ (same hand) or $\beta1 - \beta2$ (opposite hands); 90° drive: $\beta1 + \beta2 = 90°$

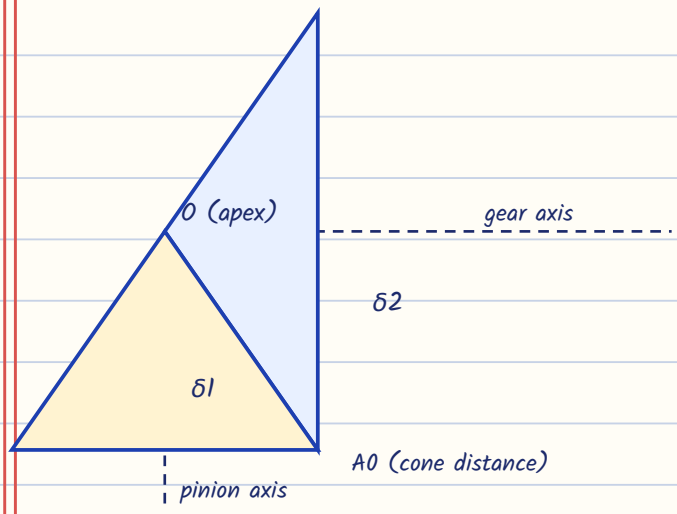
Item	Relation
Normal module	mn same for both gears
Pitch diameters	$d1 = mn\ z1 / \cos\beta1$; $d2 = mn\ z2 / \cos\beta2$
Centre distance	$a = (d1 + d2)/2 = (mn/2)(z1/\cos\beta1 + z2/\cos\beta2)$
Velocity ratio	$N1/N2 = z2/z1$ (NOT d2/d1 because β may differ); $= d2\cos\beta2 / (d1\cos\beta1)$
Efficiency	$\eta = (\cos\varphi n - \mu\tan\beta1) / (\cos\varphi n + \mu\tan\beta2)$ (for lubricated drives 90-95% at best helix angles)
Max efficiency	when $\beta1 = \beta2 = \Sigma/2$ (equal helix angles for 90° drive: 45° each)
Forces	Normal force Fn gives Ft1, Fa1 on gear 1 equal to Fa2 , Ft2 on gear 2: Ft1 = Fa2 and Fa1 = Ft2 (for 90°), plus friction component

> Design: beam strength by Lewis on virtual teeth; the limit is **wear**: permissible load Fw = k d b (k from data book); low load so it is checked for wear first.

> Materials: hardened steel on bronze or steel on cast iron; lubricate well.

III. Straight bevel gears

Bevel gears connect **intersecting shafts** (usually 90°). Pitch surfaces are **cones**. **Straight bevel**: teeth are straight and point towards the apex; cheap, used for low speed (< 5 m/s); axial thrust small. **Spiral bevel**: curved oblique teeth - smooth, high speed. **Hypoid**: shafts offset, used in automobile **differential / final drive**. **Zerol**: curved with zero spiral angle.



$\Sigma = \delta1 + \delta2$ (shaft angle, 90°)

$\tan\delta1 = z1 / z2$; $\tan\delta2 = z2 / z1$

$A0 = d1 / (2\sin\delta1) = d2 / (2\sin\delta2)$

Face width $b \le A0 / 3$ (or 10 m)

Virtual teeth $z' = z / \cos\delta$

Forces act at the **mean diameter**

$dm = d - b\sin\delta$

Kinematics and geometry

Quantity	Formula
Pitch cone angles	tan δ1 = sin Σ/(z2/z1 + cos Σ); for Σ = 90°: tan δ1 = z1/z2 , tan δ2 = z2/z1
Velocity ratio	i = N1/N2 = z2/z1 = sin δ2 / sin δ1
Cone distance	A0 = d1/(2 sin δ1) = (m/2) √(z1² + z2²) (Σ = 90°)
Face width	b ≤ A0/3 and b ≤ 10 m (take smaller; more -> poor contact at small end)
Mean pitch dia	dm = d - b sin δ ; mean module mm = m (1 - b / (2 A0))
Back cone (Tredgold)	Virtual spur gear radius Rb = r / cos δ -> **virtual teeth z' = z / cos δ**
Pitch velocity	v = π d1 N1 / 60000 (at the large end)
Addendum / dedendum	ha = m, hf = 1.2 m (Gleason: ha = 1.0 m, total depth 2.188 m)

Force analysis (Σ = 90°)

Forces are assumed to act at the **mid-point of the face width** (mean diameter dm1 = d1 - b sin δ1).

Force	Pinion	Gear
Tangential	Ft = 2T1 / dm1	Ft (equal and opposite)
Radial	Fr1 = Ft tan φ cos δ1	Fr2 = Ft tan φ cos δ2 = Fa1
Axial	Fa1 = Ft tan φ sin δ1	Fa2 = Ft tan φ sin δ2 = Fr1

> Pinion axial thrust = gear radial load, and pinion radial = gear axial (for Σ = 90°) because the axes are perpendicular.

> Axial thrust pushes the gears **apart** - bearings must locate bevel gears precisely (taper roller bearings).

> Mount bevel gears close to a bearing (avoid overhang); the shafts must be rigid.

Stresses and design of straight bevel gears

$$S_b = \sigma_b\, m\, b\, Y' (1 - b / A_0) \quad ; \quad S_w = b\, Q\, d_1\, K / \cos\delta_1$$

> Lewis on virtual teeth z' = z / cos δ, y' = 0.154 - 0.912/z'. The factor (1 - b/A0) = (A0 - b)/A0 accounts for the tapering tooth thickness (so use b ≤ A0/3).

> **Dynamic load** Fd = Ft / Cv ; Cv = 6 / (6 + v) (v = pitch-line velocity, for cut/ generated bevel gears).

> **Ratio factor** Q = 2 z2' / (z1' + z2') with virtual teeth; K = 0.16 (BHN/100)².

> Design steps: (1) materials, z1 ≥ 17 (virtual teeth), z2; (2) δ1, δ2, T; (3) assume b = A0/3 ~ 10m; (4) find m from Sb ≥ Fd; (5) check wear, give BHN; (6) calculate forces & bearings.

Example 2 – straight bevel gear pair

> P = 5 kW, N1 = 1000 rpm, i = 2, Σ = 90°, φ = 20°, σb = 140 MPa, z1 = 20, z2 = 40. T = 47,750 N·mm. δ1 = $\tan^{-1}(1/2)$ = 26.57°, δ2 = 63.43°.

> A0 = 22.36 m ; b = A0/3 = 7.45 m ; dm1 = m z1 (1 - b/2A0) = 16.67 m -> Ft = 2T/dm1 = 5729/m. z1' = 20/cos26.57° = 22.36 -> y' = 0.1132, Y' = 0.3556.

> Sb = σb m b Y'(1 - b/A0) = 140 × 7.45 m² × 0.3556 × 0.667 = 247 m² ; Fd = Ft/Cv, v = 1.047 m. m = 3: Cv = 0.656 -> Fd = 2911 N > Sb = 2223 N ✗ ; m = 4: Cv = 0.589, Ft = 1432, Fd = 2431 N ; Sb = 247 × 16 = 3952 N ✓ -> **m = 4 mm**.

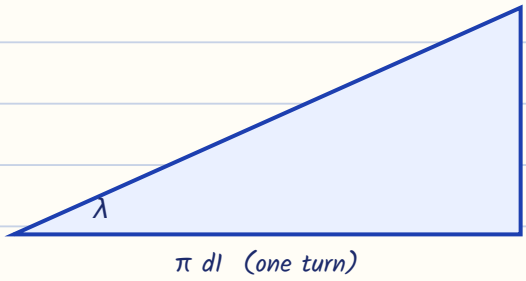
> d1 = 80, d2 = 160, A0 = 89.4, b = 30 mm. Wear: z2' = 89.4 -> Q = 1.6 ; K ≥ Fd cos δ1/(b Q d1) = 2431 × 0.894/(30 × 1.6 × 80) = 0.566 -> BHN ≥ 188. Forces: Fr1 = Fa2 = 1432 tan20° cos26.57° = 466 N ; Fa1 = Fr2 = 233 N.

IV. Worm gearing

A **worm gear drive** has a screw-like **worm** meshing with a **worm wheel**; shafts are **perpendicular, non-intersecting**. It gives a **high reduction ratio (5:1 to 100:1) in one stage**, quiet, compact, can be **self-locking**. Low efficiency (40-95%), heat generation, thrust. Uses: automobile **steering gear**, hoists, conveyors, lifts, speedometer, Torsen differential.

$\tan \lambda = L / (\pi d1)$

$= z1\ p / (\pi d1) = z1\ m / d1$



z1 = starts of worm (1, 2, 4)

q = d1 / m (diametral quotient)

i = z2 / z1

Term	Meaning / formula
Number of starts z1	Threads on the worm: 1, 2, 4 (or 6). Wheel teeth z2 ≥ 24 (30-60 typical)
Velocity ratio	i = z2 / z1 = N1 / N2
Axial pitch px	px = π m ; lead L = z1 px = π m z1
Lead angle λ	tan λ = L/(π d1) = z1 m / d1 = z1 / q

Term	Meaning / formula
Diametral quotient q	$q = d1 / m$ (8 to 17; 10 common); $d1 \approx 0.3\,a$ to $0.5\,a$
Pitch diameters	$d1 = q\,m$; $d2 = m\,z2$
Centre distance	$a = (d1 + d2)/2 = m\,(q + z2) / 2$
Face width of wheel	$b \approx 0.75\,d1$ ($z1 \le 3$); $0.67\,d1$ ($z1 > 3$); worm length $\approx 2.5\,m\,\sqrt{(z2 + 1)}$ (approx.)
Pressure angle	$\varphi n = 14.5°$ ($\lambda \le 15°$), $20°$ ($15°$-$30°$), $25°$ ($\lambda > 30°$) - bigger λ needs bigger angle

> Types: **cylindrical** (non-throated), **single-enveloping** (wheel throated - more contact), **double-enveloping / Cone (Hindley)** - highest capacity.

> **Materials**: worm = case-hardened & ground steel (20Cr/ 20MnCrS, 58-62 HRC); wheel = **phosphor bronze** / tin bronze (centrifugally cast), cast iron for light loads (bronze rim on cast-iron centre).

Forces on worm gearing and friction

Force	Without friction	With friction (coefficient μ)
Tangential on worm $Wt1$ = axial on wheel	$Wt1 = Wa1 \tan \lambda$	$Wt1 = Fn\,(\cos \varphi n \sin \lambda + \mu \cos \lambda)$
Axial on worm $Wa1$ = tangential on wheel $Wt2$	$Wa1 = 2\,T2 / d2$	$Wa1 = Fn\,(\cos \varphi n \cos \lambda - \mu \sin \lambda)$
Radial Wr (separating)	$Wr = Wa1 \tan \varphi n / \cos \lambda$	$Wr = Fn \sin \varphi n$

> Worm torque $T1 = Wt1\,d1/2$; wheel torque $T2 = Wa1\,d2/2$; note **$Wt2 = Wa1$, $Wa2 = Wt1$, $Wr2 = Wr1$**.

> **Sliding velocity** $Vs = v1 / \cos \lambda = \pi\,d1\,N1 / (60000 \cos \lambda)$ (m/s) - the whole tooth slides, causing heat.

> **Coefficient of friction** depends on Vs and materials: $\mu \approx 0.10$ at $Vs = 0.1$ m/s, 0.05 at 1 m/s, 0.03 at 5 m/s, 0.02 at ≥ 10 m/s (hard worm on bronze, oil-lubricated; use data-book curve).

Efficiency and self-locking

$$\eta = \tan \lambda\,(\cos \varphi n - \mu \tan \lambda) / (\cos \varphi n \tan \lambda + \mu) = (\cos \varphi n - \mu \tan \lambda) / (\cos \varphi n + \mu \cot \lambda)$$

> Efficiency rises with λ (up to about $45°$) and falls with μ; max at $\lambda \approx 45° - \rho'/2$ where $\tan \rho' = \mu / \cos \varphi n$ (ρ' = virtual friction angle).

> **Self-locking (irreversible)**: wheel cannot drive the worm if $\lambda < \rho'$ (i.e. $\tan\lambda < \mu/\cos\varphi_n$). Then $\eta < 50\%$. Used in hoists, lifts; no brake needed - but the worm drive efficiency is low.

> Overall efficiency of the drive also includes bearing and oil churning loss ($\eta_{total} \approx \eta \times 0.95\text{-}0.98$).

Example 3 - worm efficiency

> $z1 = 2$, $m = 5$ mm, $q = 10$ -> $d1 = 50$ mm ; $\tan\lambda = z1/q = 0.2$ -> $\lambda = 11.31°$; $\varphi_n = 20°$, $\mu = 0.05$.

> $\eta = 0.2\ (\cos20° - 0.05 \times 0.2)/(\cos20° \times 0.2 + 0.05) = 0.2 \times 0.9297/(0.1879 + 0.05) = \mathbf{78\%}$. $\rho' = \tan^{-1}(0.05/0.94) = 3.05° < \lambda$ -> not self-locking.

> Single-start worm ($\lambda = 5.71°$) with $\mu = 0.10$: $\eta = 0.1 \times 0.9297/(0.094 + 0.1) = 48\%$; $\rho' = 6.07° > \lambda$ -> **self-locking**.

Stresses in worm gear tooth, design and thermal rating

> The **wheel is weaker** (bronze) so strength is checked on the wheel: **beam strength** $Sb = \sigma b\ Cv\ b\ \pi\ m\ y$, y from Lewis for virtual teeth $z2/\cos^3\lambda$; $Cv = 6/(6 + v2)$ ($v2$ = wheel pitch velocity ≤ 3 m/s) ; σb(phosphor bronze) $\approx 80\text{-}100$ MPa.

> **Wear strength** of wheel: $Sw = d2\ b\ K_w$ (K_w = wear factor, 0.4-1.0 N/mm² by materials from data-book). Condition: $Sb, Sw \geq Fd = Wt2/Cv$.

> Worm: check bending as a beam between bearings (keep deflection small, about 0.001 d1 or less) and torsion/shear; use a thrust bearing for Wa1.

> **Thermal rating**: heat generated $Hg = P\ (1 - \eta)$ (W); heat dissipated $Hd = C\ A\ (t_{oil} - t_{air})$; $A \approx 1.14 \times 10^{-4}\ a^{1.7}$ (m^2, a in mm); $C = 8\text{-}14$ W/m²°C (natural), 20-30 with fan; $t_{oil} \leq 80\text{-}90$ °C. If $Hd < Hg$ -> fins, fan, oil cooler.

> Efficient lubrication (EP-free oil, SAE 140 / ISO VG 220-460 - sulphur additives attack bronze) and run-in are vital.

Design steps for a worm drive

1. Select z1 (from ratio and efficiency/ self-lock need), $z2 = i\ z1$, materials (hardened steel / bronze).
2. Assume centre distance a (or $d1 = a^{0.875}/2.2$) -> q, m, d1, d2; λ, φ_n; v1, Vs, μ.
3. Find $Wt2 = 2\ T2 / d2$; Fd; check wheel Sb and Sw; face width.
4. Efficiency η, power loss, thermal check; forces on bearings; worm shaft design.

V. Additional notes & worked examples

Helical gear - selection of helix angle & thrust direction

β	Effect
10° - 15°	Small thrust; modest smoothness gain; good for heavy loads with limited thrust bearing capacity
15° - 25°	**Usual range** for automobile and industrial single-helical gears
25° - 35°	Very smooth and quiet but high thrust, lower efficiency
30° - 45°	Only for double helical / herringbone gears (thrust cancels)

> **Hand rule for thrust**: for a driving RH helical gear, point the **right hand thumb along the axis**, fingers in the direction of rotation: thumb gives the thrust direction on the gear (use left hand for LH). Driven gear thrust is opposite.

> Length of contact line of helical gear is much longer than a spur gear's face width b: $l = b / \cos \beta_b$ (β_b = base helix angle) - this is the reason for higher load capacity.

> In the automobile gearbox the helix hands on the countershaft gears are chosen so that the **thrusts partly cancel**; the layout is checked for each gear selection.

Example 4 - shaft loads of the helical pair (Example 1)

> $F_t = 2244$ N, $F_r = 869$ N, $F_a = 817$ N, $d_1 = 85.1$ mm. Torque $T = 95{,}500$ N·mm.

> Bending of pinion shaft in vertical plane has an extra moment from the thrust: $M_a = F_a d_1/2 = 817 \times 42.55 =$ **34,760 N·mm**, which adds to the radial-load moment.

> If the pinion is mounted centrally between bearings 100 mm apart: bearing radial reaction (from F_t and F_r) = $\sqrt{(F_t^2 + F_r^2)}/2 = 1203$ N ; thrust 817 N taken by one (fixed) bearing -> choose angular contact / taper roller bearing for that end.

Bevel gears - types

Type	Features	Use
Straight	Teeth straight, radial to apex; simple, sudden engagement; v < 5 m/s	Hand drills, differential (small), low-speed
Spiral	Curved oblique teeth, spiral angle 35°; gradual contact, higher contact ratio, quiet, strong; thrust higher	Automobile rear axle, high-speed drives

Type	Features	Use
Zerol	Curved teeth but spiral angle 0; no extra thrust of spiral	Replace straight bevel when smoother action needed
Hypoid	Axes offset; pinion can be larger and stronger; more sliding, needs EP hypoid oil	**Car final drive**: lowers the propeller shaft (low floor)

> **Differential**: a set of bevel gears (side gears + pinions) which lets the left and right wheels rotate at different speeds in a turn while sharing torque equally; the ring gear / pinion (crown wheel & pinion) gives the final reduction (3:1 to 5:1).

> Bevel mounting: pinion on a rigid shaft with two taper roller bearings spaced apart; preload and set backlash & contact pattern with shims.

Worm gear – force and thermal examples

Example 5 – forces on a worm drive

> $P = 5$ kW, worm speed 1440 rpm, $z1 = 2$, $z2 = 40$, $m = 5$, $q = 10$ ($d1 = 50$, $d2 = 200$ mm), $\eta = 0.78$ (from Example 3). $i = 20$ -> wheel speed 72 rpm.

> $T1 = 9.55 \times 10^6 \times 5/1440 = 33{,}160$ N·mm ; $T2 = T1\, i\, \eta = 33{,}160 \times 20 \times 0.78 = 517{,}300$ N·mm.

> Wheel tangential = worm axial: $Wt2 = Wa1 = 2\,T2/d2 =$ **5173 N** ; worm tangential $Wt1 = 2\,T1/d1 =$ **1326 N** (check: $Wa1 \tan\lambda = 1035$ N frictionless, $1035/1326 = 0.78 = \eta$ ✓).

> Radial $Wr = Wa1 \tan\varphi_n / \cos\lambda = 5173 \times 0.364/0.981 = 1920$ N. Sliding velocity $Vs = \pi \times 50 \times 1440/(60000 \times 0.981) = 3.84$ m/s ($\mu \approx 0.035$-0.05 ✓).

Example 6 – thermal check of the same drive

> Heat generated $Hg = P(1 - \eta) = 5000 \times 0.22 =$ **1100 W**. Casing area $A = 1.14 \times 10^{-4}\, a^{1.7}$ with $a = (50 + 200)/2 = 125$ mm -> $A = 0.42$ m².

> Natural cooling $C = 14$ W/m²°C: $Hd = 14 \times 0.42 \times \Delta t = 5.9\,\Delta t$ -> $\Delta t = 187$ °C ✗ (oil limit about 60 °C rise).

> With fins (A doubled to 0.84 m²) and a fan ($C = 28$): $\Delta t = 1100/(28 \times 0.84) =$ **47 °C** ✓ . The oil temperature stays about 70-80 °C.

Worm wheel – construction

> Wheel rim of bronze (10–15 mm thick) bolted or shrunk on a cast-iron / steel centre to save costly bronze.

> Teeth are cut by a **hob identical to the worm** (so that contact is line contact) – the hob is fed to the correct centre distance; worm & wheel are therefore made as a matched pair.

> Run-in with light load first; after the run-in surfaces polish and efficiency improves.

> Hardness: worm hardness at least 100 BHN above the wheel; ground worm finish ≤ 0.4 µm Ra to cut friction.

VI. Comparison – helical, bevel and worm gears

Point	Helical	Bevel	Worm
Shafts	Parallel (crossed: skew)	Intersecting (90°)	Perpendicular, non-intersecting
Ratio / stage	up to 10	up to 6 (straight)	5 – 100
Efficiency	97–99%	96–98%	40–95%
Thrust	Axial $F_a = F_t \tan\beta$	Axial & radial (cone)	High axial on worm
Noise	Low	Moderate (spiral low)	Very low
Self-locking	No	No	Possible ($\lambda < \rho'$)
Automobile use	Gearbox	Differential, steering	Steering, speedometer

VII. Quick revision & important questions

Formulas to remember

> Helical: $m_t = m_n/\cos\beta$; $a = m_n(z_1 + z_2)/(2\cos\beta)$; $z' = z/\cos^3\beta$; $F_a = F_t \tan\beta$; $F_r = F_t \tan\varphi_n/\cos\beta$; $p_x = \pi m_n/\sin\beta$.

> Bevel: $\tan\delta_1 = z_1/z_2$; $A_0 = d_1/(2\sin\delta_1)$; $b \le A_0/3$; $F_{r1} = F_t \tan\varphi \cos\delta_1$; $F_{a1} = F_t \tan\varphi \sin\delta_1$; $z' = z/\cos\delta$.

> Worm: $\tan\lambda = z_1/q$; $a = m(q + z_2)/2$; $V_s = v/\cos\lambda$; $\eta = (\cos\varphi_n - \mu\tan\lambda)/(\cos\varphi_n + \mu\cot\lambda)$; self-lock $\lambda < \rho'$.

> Strength: $S_b = \sigma_b\, m_n\, b\, Y'$; $S_w = b\, Q\, d_1\, K/\cos^2\beta$ (helical) ; S_b(bevel) has $(1 - b/A_0)$.

1. Explain the kinematics and tooth proportions of a parallel helical gear. (5 M)
2. Derive the forces acting on a helical gear tooth. (5 M)
3. Design a helical gear pair for given power, speed and ratio. (10 M)
4. Write on crossed helical gears: relations and efficiency. (5 M)
5. Derive force components and stresses of a straight bevel gear. (10 M)
6. Design a pair of straight bevel gears for given data. (10 M)
7. Explain worm gearing: kinematics, forces, friction & efficiency, self-locking. (10 M)
8. A worm drive transmits P kW - find efficiency, stresses and the heat to be dissipated. (10 M)

UNIT IV

Design of Gear Boxes

speed reducers | multi-speed gear boxes for machine tools | structural diagrams | ray diagrams | automobile gearbox

I. Introduction

A **gear box** is a closed housing containing gears, shafts, bearings and lubricant that **changes speed and torque** between a prime mover and a machine. Two major types:

Type	Purpose	Examples
Speed reducer (fixed ratio)	Reduces motor speed to the machine speed with a constant ratio and increases torque	Conveyors, mixers, cranes, hoists, mills
Multi-speed gear box	Gives a **number of different speeds** (stepped) from a constant-speed motor	Machine tool headstocks (lathe, milling, drilling)
Automobile gear box	Selects torque multiplication for starting / hill / cruising; reverse & neutral	Sliding-mesh, constant-mesh, synchromesh, automatic

II. Design of speed reducers

Gear type	Ratio per stage	Efficiency	Remark
Spur / helical	4 - 6 (spur), 6 - 10 (helical)	97-99%	Parallel shafts; helical quieter, thrust
Bevel (straight / spiral)	up to 5 - 6	96-98%	Right-angle drive; often first stage
Worm	5 - 60 (single stage)	50-92%	Right angle, compact, silent; self-locking possible
Planetary (epicyclic)	3 - 10 per stage	97-98%	Coaxial, compact, high torque density

Types of reducers by number of stages

> **Single reduction**: one gear pair - $i \leq 6$-8. **Double reduction**: two pairs - i up to 30-40 (most common). **Triple**: i up to 100+.

> **Layouts**: in-line (coaxial) - input and output shafts on the same axis; **unfolded** (all shafts in one plane); **bevel-helical** (right angle, first bevel then helical); **worm-helical**.

> **Splitting of ratio**: for a double reduction, $i = i1 \times i2$ with **$i1 = 1.2$ to $1.4 \times i2$** (first, high-speed stage has smaller torque so can take the larger ratio; helps in oil-dip lubrication of both gears). For three stages take $i1 > i2 > i3$ (each about 1.2-1.3 times the next), so the low-speed gears are not oversized.

> Minimum centre distance / volume is found when each stage is designed for **equal strength and wear**, which gives $i1/i2 \approx 1.3$ for typical conditions.

Design procedure for a speed reducer

1. Collect: motor power P, speed N1, output speed N2 (overall $i = N1/N2$), duty, service factor, space.
2. Decide the number of stages & the **ratio split**; determine speeds & torques of each shaft with efficiencies (0.97-0.98 per stage including bearings):

$$T_{out} = 9.55 \times 10^6 \, P \, \eta / N_{out} \ \ (N{\cdot}mm); \quad \eta_{overall} = \eta_{gear}^{n} \times \eta_{bearing}^{m}$$

3. Design the **gear pairs** (Unit II/III): material, module, teeth, face width; check bending and wear for each stage.
4. Design **shafts** for torque + bending (gear forces Ft, Fr, Fa) using ASME code; select keys / splines.
5. Select **bearings** for each shaft from L10h life (e.g. 10,000-20,000 h industrial; 3,000-5,000 h automotive).
6. Design the **housing**, covers, lubrication, seals, breathers; check **heat dissipation**.
7. Prepare layout: gear/bearing positions, shaft steps, shoulders and circlips; assembly dimensions.

Example 1 - shaft speeds & torques of a double reducer

> Motor 7.5 kW at 1440 rpm; output about 90 rpm -> $i = 16$. Split: $i1 = 1.25\ i2$ -> $i2 = \sqrt{(16/1.25)} = 3.58$, $i1 = 4.47$. Efficiency 0.98 per stage.

> Shaft I: $N = 1440$ rpm, $T1 = 9.55 \times 10^6 \times 7.5/1440 = 49{,}740$ N·mm.

> Shaft II: $N = 1440/4.47 = 322$ rpm, $T2 = T1 \times i1 \times 0.98 =$ **217,900 N·mm**. Shaft III: $N = 322/3.58 = 90$ rpm, $T3 = T2 \times i2 \times 0.98 =$ **764,400 N·mm**.

> Output power = $T3\ N/9.55 \times 10^6 = 7.2$ kW (overall $\eta = 0.98^2 = 0.96$ ✓). Each stage gear pair is now designed from T1, T2 (Unit II/III method).

Example 1 (contd.) – first-stage gear and bearing on shaft 1

- Stage 1: z1 = 20, z2 = 89 (i1 = 4.45), 20° FD, C45, σb = 140 MPa, Cs = 1.25, ordinary cut gears. Design torque 49,740 × 1.25 = 62,175 N·mm; y = 0.1084; b = 10 m.
- m^3 Cv = 2T/[z1 σb (b/m) π y] = 124,350/[20 × 140 × 10 × π × 0.1084] = 13.0. m = 3: v = 4.52 m/s, Cv = 0.399, m^3Cv = 10.8 ✗ ; m = 4: v = 6.03, Cv = 0.332, m^3Cv = 21.2 ✓ -> **m = 4 mm**.
- d1 = 80, d2 = 356 mm, a = 218 mm, b = 40 mm. Ft = 2 × 49,740/80 = 1243 N ; Fd = Cs Ft/Cv = 4682 N ; Sb = 140 × 4 × 40 × 0.3405 = 7627 N ✓. Wear: Q = 1.63 -> K ≥ 4682/(40 × 1.63 × 80) = 0.90 -> **BHN ≥ 237** (take 240).
- Housing wall δ = 0.025 × 218 + 3 = 8.5 -> 9 mm. Bearing on shaft 1: Fr = Ft tan20° = 452 N, resultant 1323 N, central pinion -> 660 N per bearing; life 20,000 h at 1440 rpm: C = P (60 n L10h/10^6)^(1/3) = 0.66 × 12 = 7.9 kN -> with shock 1.3: 10.3 kN -> choose a 6205 deep-groove (C about 14 kN, check catalogue).

Shaft design (ASME code) in the gear box

- d^3 = (16/(π τ_allow)) √[$(K_b M)^2 + (K_t T)^2$] with τ_allow = 0.18 σut or 0.3 σy (lower; 25% less if keyway), Kb = 1.5–2 (suddenly applied, minor shock), Kt = 1–1.5.
- M = resultant bending moment from gear forces in two planes: M = √($M_h^2 + M_v^2$); axial thrust adds a moment Fa d/2 (helical, bevel, worm).
- Check deflection (< 0.0005 rad at gear) and critical speed; use standard sizes & fillet radius ≥ 0.1 d to cut stress concentration.
- Materials: C45 / 40C8 normalised or 40Cr1, 20MnCr5 for pinion shafts (integral with gear), ground fits h6/k6 at bearings.

Housing, lubrication & accessories

Item	Practice
Housing material	Grey cast iron FG200/FG260 (damping, cheap); welded steel for single piece; aluminium for light, small gearboxes.
Wall thickness	δ ≈ 0.025 a + 3 mm (a = centre distance, min 8 mm); ribs around bearing bosses; split along shaft axes for assembly.
Lubrication	**Splash / oil bath** for gear pitch velocity v < 12 m/s (oil level one tooth height for the lowest gear); **forced spray** for v > 12 m/s with pump, filter, cooler.

Item	Practice
Oil grade	ISO VG 150-320 (EP) for industrial reducers; SAE 80W-90 GL-4 for automobile gears; hypoid GL-5.
Accessories	**Breather** (pressure relief), oil level gauge, drain plug (magnetic), filler / inspection cover, lifting lugs, dowel pins, gaskets, oil seals.
Sealing	Lip seals, labyrinth, V-ring, O-ring on covers; bearing end caps with shims for end float.
Thermal	Heat generated Hg = P (1 - η); dissipated Hd = C A (t_oil - t_a) with C = 12-18 W/m²°C (natural), A = housing area; t_oil ≤ 80-90 °C; use fins/fan/oil cooler.

III. Multi-speed gear boxes for machine tools

A machine tool (lathe, drilling, milling) needs **different spindle speeds** for different cutting speed/ diameter/ tool material, from a constant-speed motor. The multi-speed gear box gives **z stepped speeds** N1 < N2 < ... < Nz between Nmin and Nmax.

Choice of progression of spindle speeds

Series	Law	Remark
Geometric progression (GP)	N2 = φ N1, N3 = φ² N1, ... Nk = N1 φ^(k-1)	**Most used**: constant % loss of cutting speed between steps; fewer gears; uses preferred numbers
Arithmetic progression (AP)	N_k = N1 + (k - 1) d	Too many closely spaced high speeds; wasteful of time (not used)
Harmonic progression (HP)	1/N in AP	Used in some special / feed boxes
Logarithmic	log N in AP	GP is the same as logarithmic progression

> **Speed range** Rn = Nmax/Nmin = φ^(z-1) -> **φ = (Nmax/Nmin)^(1/(z-1))** ; z = 1 + log Rn / log φ.

> **Percentage loss of cutting speed** (maximum, just above a step) = (φ - 1)/φ × 100. Eg φ = 1.26 ->

20.6%, 1.41 -> 29%, 1.12 -> 10.7%.

> Why GP? The economic loss in cutting speed is the **same percentage** at every speed step -> best compromise between number of steps and loss.

Standard ratio φ (preferred number series)	Series	Ratio	Max speed loss
R5	10^(1/5)	φ = 1.58	36.7%
R10	10^(1/10)	**φ = 1.26**	20.6%
R20	10^(1/20)	**φ = 1.12**	10.7%
R20/3	(R20 every 3rd)	**φ = 1.41**	29%
R40	10^(1/40)	φ = 1.06	5.7%
R10/2	(R10 every 2nd)	φ = 1.58	36.7%

Standard spindle speeds (R10, φ = 1.26) in rpm

> 31.5, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000 … (IS 1076 preferred numbers; every 10th term is 10 × the previous decade).

> Eg. 16-speed lathe: Nmin = 31.5, Nmax = 1000 -> Rn = 31.7, φ = 31.7^(1/15) = 1.26 ✓ (R10).

IV. Kinematic layout of the gear box

> **Sliding gear clusters** (cones of 2 or 3 gears) on splined shafts give the stepped ratios; only **one pair** of a cluster is in mesh at a time. Also clutches and Norton tumblers (feed boxes).

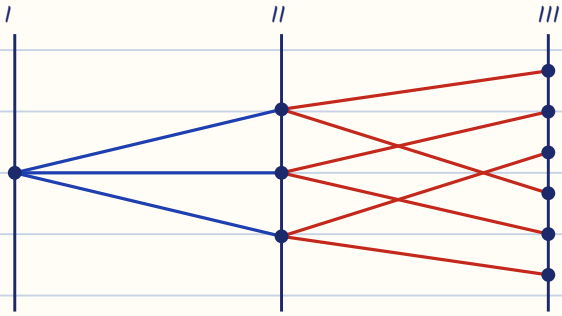
> Number of speeds z = p1 × p2 × p3 … where p = number of gear pairs (transmissions) in a group; p = 2, 3 (4 only with care). Eg 12 = 3 × 2 × 2.

> Total gear pairs = p1 + p2 + p3 …, shafts = number of groups + 1 ; gears = 2 × (p1 + p2 + …).

> Constant centre distance in each group: all pairs have the **same sum of teeth** Zs = z_driver + z_driven (same module).

Structural formula and structural diagram

Structural formula: z = p1 (x1) · p2 (x2) · p3 (x3) … where the number in brackets is the **characteristic x** of each group = **number of steps (in terms of φ) between adjacent ratios of that group**.



Structural formula

z = 3(1) × 2(3)

Shows only the **pattern** of ratios (no speed values), equal spacing.

Rule / quantity	Meaning
Normal ('basic') structure	x1 = 1 ; x2 = p1 ; x3 = p1 p2 ; ... (all speeds distinct, no overlap)
Range of a group	$R_g = \varphi^{(x(p-1))}$
Limits	Ratio of any gear pair: **1/4 ≤ i ≤ 2** (min 1/4: reduces size & noise; max 2 : avoids high speeds on spindle)
Maximum group range	$R_g \leq 8$ i.e. $\varphi^{(x(p-1))} \leq 8$ (ratio 4 : 1/2)
Ordering of groups	Characteristics generally increase toward the spindle; last group has the largest range -> intermediate shafts rotate fast & torque is small (smaller gears)
Choice of structure	Smallest weight/ size: reduce speeds **late** (speeds high on intermediate shafts, low torque)

z	Structural formulas (possible)	Group ranges ($\varphi^{x(p-1)}$)
4	2(1)·2(2)	φ ; φ^2
6	**3(1)·2(3)** ; 2(1)·3(2) ; 2(3)·3(1)	φ^2 ; φ^3
8	2(1)·2(2)·2(4)	φ ; φ^2 ; φ^4
9	3(1)·3(3)	φ^2 ; φ^6
12	**3(1)·2(3)·2(6)** ; 2(1)·3(2)·2(6) ; 2(1)·2(2)·3(4)	$\varphi^2,\varphi^3,\varphi^6$; $\varphi,\varphi^4,\varphi^6$; $\varphi,\varphi^2,\varphi^8$
16	2(1)·2(2)·2(4)·2(8)	φ ; φ^2 ; φ^4 ; φ^8

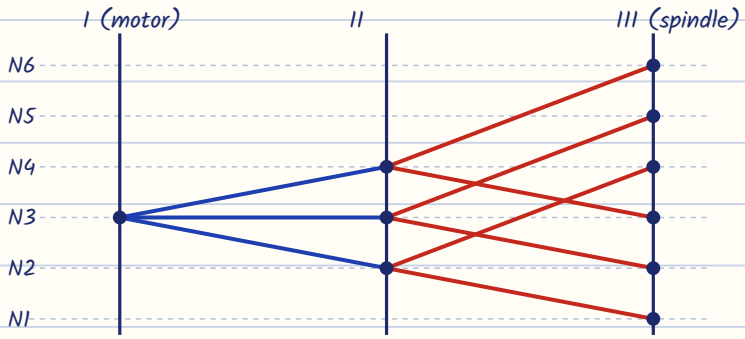
> Check: for φ = 1.26 (R10) 3(1)·2(3)·2(6) -> group 3 range $\varphi^6 = 4$ ✓ ≤ 8. For φ = 1.41, $\varphi^6 = 8$ ✓ borderline; $\varphi^8 = 16$ ✗ so 2(1)·2(2)·3(4) is not allowed for φ = 1.41.

> A **structural diagram** shows the pattern with equally spaced nodes (not to speed scale); a **ray**

(speed) diagram puts the shafts at actual speed levels (log scale).

V. Ray diagram

A **ray diagram** (speed diagram) is a graph with **vertical lines for shafts** and **horizontal lines for speeds on a logarithmic scale**, equally spaced by log φ. Each ray joins the speed of the driver to the speed of the driven shaft: **rising ray** = speed increase ($i > 1$), **falling ray** = reduction ($i < 1$), **horizontal** = $i = 1$. The number of steps up/down = log i / log φ.



Ray diagram (6 speeds, φ = 1.41)

Vertical lines = shafts;

horizontal lines = speeds on a log scale (equal spacing = φ).

Sloping up = speed ↑ ($i > 1$),

down = speed ↓ ($i < 1$).

Blue: first group (3 ratios, step φ^1).

Red: second group (2 ratios, step φ^3).

Rules for drawing ray diagrams

1. Draw as many vertical lines as shafts (motor, I, II, ..., spindle) and horizontal lines equal to the number of speeds, with spacing log φ.
2. Mark the spindle speeds N1 .. Nz on the last vertical line; mark the motor (or belt-reduced) speed on the first.
3. Start from the first group with the number of rays = p1, then the next group, so that all z spindle speeds appear (use the structural formula).
4. Ray limits: a rising ray may not exceed $i = 2$ (φ = 1.41: 2 steps up; φ = 1.26: 3 steps up); a falling ray may not go below $i = 1/4$ (φ = 1.41: 4 steps down; φ = 1.26: 6 steps down).
5. Keep intermediate-shaft speeds as **high as possible**; and the spindle's last transmission can be the largest reduction.

Example 2 – 6-speed gear box (spindle 100 – 560 rpm, motor via belt to 200 rpm)

> z = 6, Nmin = 100, Nmax = 560 -> Rn = 5.6, φ = 5.6^(1/5) = 1.41 (R20/3). Speeds: **100, 140, 200, 280, 400, 560 rpm**. Structural formula **3(1)·2(3)**.

> Group 1 (3 ratios, x = 1): ratios 1/φ, 1, φ -> shaft II speeds 140, 200, 280. Group 2 (2 ratios, x = 3): ratios 1/φ and φ² (= 2) -> from each shaft-II speed to a lower (÷1.41) and a higher (× 2) spindle speed.

> Check limits: ratios 0.71, 1, 1.41, 2 all lie between 1/4 and 2 ✓ ; group 2 range = φ³ = 2.8 ≤ 8 ✓.

Determination of number of teeth

1. For a group with sum 2s, a ratio i = a/b (as a ratio of coprime integers) gives driver z = 2s a/(a + b) ; driven 2s - z.
2. Choose 2s such that all driver/driven teeth are integers ≥ **17–20** (minimum on the pinion), and ≤ 100–120; ratios must match φ-series within ± 10 (φ - 1) %.
3. Check actual output speeds, tooth strength (design the smallest pinion of each group) and module (same m in the group).

Teeth for Example 2

> Group 1: 2s = 72: i = 0.707 -> 30/42 (0.714); i = 1 -> 36/36; i = 1.41 -> 42/30 (1.40). Group 2: 2s = 84: i = 0.707 -> 35/49 (0.714); i = 2 -> 56/28 (2.0).

> Speeds: shaft II = 200 × (0.714, 1, 1.4) = 143, 200, 280; spindle = {143 × 0.714, 143 × 2, 200 × 0.714, 200 × 2, 280 × 0.714, 280 × 2} = **102, 143, 200, 286, 400, 560 rpm** (errors ≤ 2% < 4.1% ✓).

> Gears total = 2 (3 + 2) = 10; shafts = 3 (I, II, spindle).

Example 3 – 12-speed gear box (R10, φ = 1.26)

Spindle speeds 40 to 500 rpm: **40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500** (Rn = 12.5; φ = 12.5^(1/11) = 1.26). Structural formula **3(1)·2(3)·2(6)**. Input shaft I = 200 rpm (motor 1440 rpm through a belt drive).

Group	p (x)	Ratios (steps)	Numeric ratios	2s	Teeth driver / driven
I (I -> II)	3 (1)	φ^{-3}, φ^{-2}, φ^{-1}	0.50, 0.63, 0.79	72	24/48, 28/44, 32/40
2 (II -> III)	2 (3)	φ^{-1}, φ^{2}	0.79, 1.59	80	35/45, 49/31

Group	p (x)	Ratios (steps)	Numeric ratios	2s	Teeth driver / driven
3 (III -> spindle)	2 (6)	φ^{-3}, φ^{3}	0.50, 2.00	90	30/60, 60/30

> Speeds: shaft II = 100, 127, 160 ; shaft III = 78, 99, 124, 158, 201, 253 ; spindle = 39, 49.5, 62, 79, 100.5, 126, 156, 198, 249, 316, 402, 506 rpm - errors ≤ 2.8% (refine teeth to match ± 2.6% if required).

> Number of gears = 2 (3 + 2 + 2) = 14; shafts = 4. Group ranges: φ^{2} = 1.6, φ^{3} = 2, φ^{6} = 4 (all ≤ 8 ✓). All ratios between 0.5 and 2 ✓ ⊂ [1/4, 2].

> A 9-speed box is 3(1)·3(3): ranges φ^{2} and φ^{6} -> with φ = 1.41 the last group range is φ^{6} = 8 (limit).

Layout & design notes for machine-tool gear boxes

> **Sliding cluster**: gears in one cluster must have a difference of at least 4 teeth between adjacent gears to avoid fouling; axial width kept small; shifting fork at the groove.

> Use **two-gear clusters** (2 pairs) where possible: shorter shaft & housing; 3-gear cluster needs interlocking so two clusters cannot be engaged at once.

> Module is normally the **same for a group** (the same 2s), but can differ between groups: higher torque near the spindle -> larger module.

> **Shaft diameters** are found from the torque at the lowest speed with ASME formula; intermediate shaft diameter < spindle diameter; spindle of lathe has bore for bar and is stiff (deflection limit).

> **Bearings**: spindle has precision tapered roller or angular contact pair (preloaded); others ball bearings; lubrication by oil bath/ splash, with filter & sight glass.

> Efficiency: each gear pair about 0.97-0.98, each bearing pair 0.99; overall about 0.9 for a 3-group box.

Planetary (epicyclic) speed reducer

> Elements: sun gear, planets on a carrier, ring (annulus). Ratio (ring fixed, sun input, carrier output) = 1 + z_ring/z_sun = 3 to 10; coaxial input/output, several planets share the load (3-4) so power density is high.

> **Tabular method**: with the carrier fixed, rotate gears; then add the carrier's rotation. Used in automatic gearboxes, wheel hubs, winches.

> Assembly condition: (z_sun + z_ring)/n_planets = integer; z_ring = z_sun + 2 z_planet.

Gear box faults & maintenance

Symptom	Likely cause	Remedy
Noise / whine	Wrong backlash, worn or misaligned gears, low oil	Re-shim, replace, correct oil level
Overheating	Overload, too much or too little oil, wrong viscosity, blocked breather	Check load, oil grade/level, clean breather, add cooler
Oil leakage	Worn seals, loose cover bolts, clogged breather pressure	Replace seal/gasket, clean breather
Hard gear shifting	Worn synchroniser, clutch drag, wrong oil	Replace synchro, adjust clutch
Bearing failure	Contaminated oil, wrong preload/fit	Filter oil, correct fits, replace bearings

VI. Design of a machine-tool gear box – summary of steps

1. Fix Nmin, Nmax, z (from the work range), choose φ from the standard series; list spindle speeds.
2. Choose the **structural formula**; check group ranges (≤ 8) and ratios ($1/4 \le i \le 2$).
3. Draw **ray diagram**; read all gear ratios.
4. Select sums of teeth Zs and find tooth numbers; compute actual speeds and errors.
5. Compute power/torque on each shaft (lowest speed of a shaft governs the maximum torque), design the **weakest pinion** of each group for bending & wear (Units II-III), face width b = 10 m.
6. Design shafts (torque at the lowest speed), splines, keys, bearings, clutch/ cluster shifting, lubrication, housing.

> Power at the spindle is kept constant (constant-power drive) between steps: torque = P/ω is greatest at the **lowest spindle speed**.

> Gear-shifting: sliding cluster on splines with fork; interlock; **dog clutch / friction clutch** for fast change; in modern machines VFD (inverter) motors reduce steps.

VII. Automobile gear box (extra – relevant to the subject)

Type	Working	Features
Sliding mesh	Gears on splined shaft slide to engage countershaft gears	Simple, noisy, needs double declutching; spur / helical; old trucks
Constant mesh	All gears always in mesh; dog clutches select the gear	Helical gears, quieter, easier shift

Type	Working	Features
Synchromesh	Cone synchronisers equalise speeds before dog engagement	Smooth shift without double clutching; modern cars
Epicyclic / automatic	Planetary gear sets with clutches and bands (or CVT / DCT)	Torque converter, auto shifting

Choice of gear ratios in a vehicle gearbox

> **Top gear ratio** (overall) from maximum speed: N_engine at max power -> vehicle speed V = 0.377 N r_w/(G_total) (km/h, r_w in m).

> **First gear ratio** from **gradeability** / starting: engine torque × G1 × G_final × η / r_w ≥ W (sin α + f cos α); usually G1 = 3.5 - 4.5, reverse about 3.5 - 4.

> **Intermediate gears in GP**: G_k = G1 (G_top/G1)^((k-1)/(n-1)) - gives equal % speed change on shifts. Eg 4-speed G1 = 4, top = 1: G = 4, 2.52, 1.59, 1.0.

> Practice: ratios are slightly 'closer' (progressive) at high gears for better acceleration; 5th gear is an overdrive (0.8).

> Gearbox rated at maximum engine torque × G1 ; helical gears 20° normal, β = 20°-30°; gear material 20MnCrS case-hardened, shaved; shafts splined; bearing life about 3000-5000 h.

VIII. Quick revision & important questions

Remember

> Speed reducer: i1 ≈ 1.3 i2 for double reduction; T_out = 9.55 × 10^6 P η/N.

> GP: N_k = N1 φ^(k-1); φ = (Nmax/Nmin)^(1/(z-1)); standard φ: 1.12, 1.26, 1.41, 1.58, 2.

> Structural formula z = p1(x1)·p2(x2) ...; x1 = 1, x2 = p1, x3 = p1 p2; group range φ^(x(p-1)) ≤ 8; ratio 1/4 ≤ i ≤ 2.

> Ray diagram: shafts vertical, speeds horizontal (log scale). Teeth from sum Zs and ratio a/b.

1. Explain the design procedure of a two-stage speed reducer. (10 M)
2. Why geometric progression is used for spindle speeds? Derive φ. (5 M)
3. Explain structural formula and the rules for ray diagrams. (10 M)
4. Design a 9-speed (or 12-speed) gear box for a lathe: structural & ray diagrams, teeth of gears. (10 M)

5. Determine the number of teeth on each gear of a 6-speed gear box. (10 M)
6. Explain the types of automobile gear boxes with neat sketches. (5 M)

UNIT V

Elements of Motion Control: Clutches & Brakes

internal expanding | external contracting | band | cone | energy considerations | temperature rise | friction materials

I. Introduction

A **clutch** connects / disconnects two **rotating** shafts (engine and gearbox) so that power can be transmitted or interrupted smoothly. A **brake** **slows down or stops** a rotating body by absorbing its kinetic energy as heat, or holds it at rest. Both work on **friction** between a rotating member and a stationary (brake) or another rotating (clutch) member.

Type	Clutch	Brake
Positive / mechanical	Jaw (dog) clutch, spline clutch - no slip, engage at rest or low speed	Ratchet and pawl
Friction	Disc / plate, cone, centrifugal, band, internal expanding, external contracting	Drum (internal expanding), external shoe, band, disc
Other	Hydraulic (fluid coupling), electromagnetic, one-way (sprag)	Eddy-current, regenerative, engine brake

> Automobile: **single dry plate clutch** (engine-gearbox), **drum brake** at rear (internal expanding shoes), **disc brake** at front, **band brake** in automatic transmissions, **centrifugal clutch** in scooters / mopeds, **cone clutch** in synchronisers.

> Requirements: high and **steady** μ, good heat dissipation & high temperature strength, smooth engagement, long wear life, easy adjustment.

II. Friction materials

Material (against CI / steel)	μ (dry)	Max pressure (MPa)	Max temp (°C)	Remarks
Organic / resin-bonded (non-asbestos, moulded or woven)	0.35 - 0.40	0.7 - 1.0	200 - 260	Most automotive linings; quiet, soft; fades at high temp

Material (against CI / steel)	μ (dry)	Max pressure (MPa)	Max temp (°C)	Remarks
Semi-metallic (steel wool + resin)	0.35 - 0.45	1.0 - 1.4	300 - 450	Heavy-duty, good conduction; noisy, wears disc
Sintered metal (Cu / Fe base)	0.25 - 0.40	1.5 - 3.0	400 - 600	High load, high temp, racing, heavy clutches; abrasive
Ceramic / carbon-carbon	0.30 - 0.45	> 1.0	600+	Light, stable, expensive (sports, aircraft)
Cork (in oil)	0.30 - 0.35	0.05 - 0.10	80	Wet clutches, soft engagement
Leather / cotton / wood	0.3 - 0.5	0.07 - 0.3	60 - 100	Old, belt clutches, light duty
Cast iron on cast iron (dry)	0.15 - 0.20	0.1 - 0.3	300	Brake drums, disc rotors (the rubbing partner)
Asbestos (old; now banned)	0.35 - 0.45	0.5 - 1.0	350	Banned - health hazard (asbestosis)

> Values are typical - use the data-book value for the problem. Wet (oil) μ is 0.05-0.15 lower than dry.

> **Properties wanted**: high coefficient of friction, small change with temperature (**fade resistance**), good wear resistance, thermal conductivity, strength, low compressibility, no noise/ smell, low cost, environmentally safe (no asbestos).

> **Rubbing partner**: grey cast iron (FG260), nodular iron, steel (hardened) - cast iron has good damping & heat capacity.

> **Design pressure** p_{max} is chosen from the table to give a reasonable wear life (pressure × velocity limit: $p\ v \leq$ allowable).

III. Pressure distribution and torque capacity

For a friction surface the pressure distribution may be assumed **uniform pressure** (new surfaces) or **uniform wear** (after run-in: $p\ r$ = constant). Uniform-wear (the more realistic design basis) gives a lower torque capacity, so it is safer.

	Uniform pressure	Uniform wear
Axial force W (disc, radii R1 outer, R2 inner)	$W = p\ \pi\ (R1^2 - R2^2)$	$W = 2\pi\ p_{max}\ R2\ (R1 - R2)$ ($p\ r = C = p_{max}\ R2$)

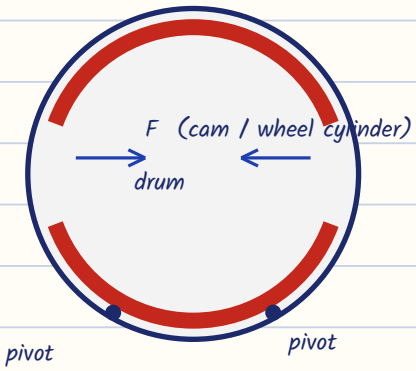
	Uniform pressure	Uniform wear
Torque per friction surface	$T = (2/3)\ \mu\ W\ (R1^3 - R2^3)/(R1^2 - R2^2)$	$T = \mu\ W\ (R1 + R2)/2 = \mu\ W\ R_m$
Mean radius R_m	$(2/3)(R1^3 - R2^3)/(R1^2 - R2^2)$	$(R1 + R2)/2$

> **Plate (disc) clutch**: torque $T = n\ \mu\ W\ R_m$ where n = number of friction surface pairs (single plate has 2 surfaces: $n = 2$).

> Typical car clutch: $R1/R2 \approx 1.2\text{-}1.3$; springs give W; $\mu \approx 0.3\text{-}0.4$; torque capacity = (1.2 - 1.5) × max engine torque (to avoid slipping).

IV. Internal expanding rim clutches and brakes

Two (or more) **shoes** lined with friction material are **expanded** outwards (by cam / wheel cylinder / centrifugal force) against the **inner surface of a rotating drum**. Used as **automobile drum brakes** and **centrifugal clutches**. The shoes are pivoted at one end (anchor), the other end is pushed by the actuating force F.



Leading shoe: friction helps to press the shoe (self-energising)

Trailing shoe: friction opposes (self-de-energising)

Torque $T = \mu\ pa\ b\ r^2\ (\cos\theta 1 - \cos\theta 2)/\sin\theta a$

b = shoe width, r = drum radius

Analysis of a pivoted shoe

> Notation: r = drum radius, b = shoe width (face), $\theta 1$, $\theta 2$ = angles of shoe start & end measured from the pivot line, a = distance of pivot from drum centre, c = distance of F from pivot (as in the figure), θa = angle at which the pressure is maximum: $\theta a = 90°$ if $\theta 2 \geq 90°$, else $\theta a = \theta 2$.

> Pressure at any point is proportional to the radial displacement of the lining, which is proportional to $\sin\theta$: **$p = pa \sin\theta / \sin\theta a$** ($pa$ = maximum pressure).

Quantity	Formula
Brake torque T (per shoe)	$T = \mu\ pa\ b\ r^2\ (\cos\theta 1 - \cos\theta 2) / \sin\theta a$
Moment of normal forces about pivot	$Mn = (pa\ b\ r\ a / \sin\theta a)\ [\ (\theta 2 - \theta 1)/2 - (\sin 2\theta 2 - \sin 2\theta 1)/4\ ]$
Moment of friction forces about pivot	$Mf = (\mu\ pa\ b\ r / \sin\theta a)\ [\ r\ (\cos\theta 1 - \cos\theta 2) - a\ (\sin^2\theta 2 - \sin^2\theta 1)/2\ ]$
Actuating force F	Leading (self-energising) shoe: $F = (Mn - Mf)/c$; trailing (de-energising) shoe: $F = (Mn + Mf)/c$
Self-locking	when $Mn \leq Mf$ ($F \leq 0$): brake grabs on its own. Avoid by designing $Mn > Mf$ (smaller a, μ)
Pivot reactions	$Rx = \int$ components of p and friction; resultant $= \sqrt{(Rx^2 + Ry^2)}$ for design of the anchor pin
Zero-friction-moment pivot (symmetrical shoe)	$a = 4\ r \sin\alpha / (2\alpha + \sin 2\alpha)$ (α = half angle of shoe) -> $Mf = 0$, wear even, no self-energising

> **Leading shoe**: friction drags the shoe into the drum (self-energising, needs less F, wears more, higher pressure). **Trailing shoe**: friction pulls it away. For the same F the leading shoe gives more torque - a duplex / 'two leading shoe' brake uses both leading in forward motion.

> Total brake torque = T(leading) + T(trailing), with the pressure pa of the trailing shoe less than that of the leading shoe for the same F.

> Heat dissipation: drum brake dissipates heat through the drum only; fade at high temperature is the main limit.

Example 1 – internal expanding brake (two shoes)

> Drum radius r = 150 mm, shoe width b = 40 mm, θ1 = 10°, θ2 = 120° (so θa = 90°), pivot distance a = 150 mm, μ = 0.3, F acts at c = 250 mm, leading shoe pa = 0.8 MPa.

> Torque (leading) $T = \mu\, pa\, b\, r^2(\cos 10° - \cos 120°) = 0.3 \times 0.8 \times 40 \times 150^2 \times 1.4848 =$ **320,700 N·mm**.

> $Mn = pa\, b\, r\, a\, [\,(\theta 2 - \theta 1)/2 - (\sin 240° - \sin 20°)/4\,] = 0.8 \times 40 \times 150 \times 150 \times 1.262 = 908{,}600$ N·mm ; $Mf = 0.3 \times 0.8 \times 40 \times 150\, [150 \times 1.4848 - 150 \times (0.75 - 0.0302)/2] = 243{,}000$ N·mm.

> $F = (Mn - Mf)/c = (908{,}600 - 243{,}000)/250 =$ **2662 N**. Trailing shoe with the same F: $pa' = F\, c/((Mn + Mf)/pa) = 0.462$ MPa -> $T' = 185{,}400$ N·mm. Total brake torque = 320,700 + 185,400 = **506,100 N·mm ≈ 506 N·m**.

Centrifugal clutch (internal expanding clutch)

> Shoes carried on a spider fixed to the driving shaft; they fly outward against a drum of the driven member when the speed exceeds the **engagement speed** (about 0.7 of running speed), held back by garter springs.

> Centrifugal force per shoe $Fc = m\, \omega^2\, r_g$; spring force $Fs = m\, \omega 1^2\, r_g'$ at engagement; net normal force on drum $R_N = Fc - Fs$.

> Torque transmitted $T = n\, \mu\, R_N\, r$ (n = number of shoes). Used in scooters, mopeds, starters, chain saws - automatic, no pedal.

> Design of drum / shoes: from the power: $T = 9.55 \times 10^6\, P/N$; contact pressure $p = R_N / (b\, l) \leq$ allowable (0.1 MPa for cork/ leather, 0.3 MPa for asbestos-free).

V. External contracting rim clutches and brakes

Shoes press against the **outside** of the rotating drum (contracting). Types: **single block (short shoe)**, **pivoted block (long shoe)**, **double block** (two blocks acting on the drum from opposite sides - balances the shaft load) and the pivoted long shoe is most used in industry / railways.

1. Single block (short shoe) brake

> Shoe subtends a small angle (< 45°) so the pressure is almost uniform; normal force R_N on the drum, friction force $Ft = \mu\, R_N$; torque $T_B = \mu\, R_N\, r$.

> Lever: force F at distance l from the fulcrum, shoe at distance x along the lever, line of the friction force at a perpendicular distance a from the fulcrum:

Case	Actuating force F
Friction force through fulcrum (a = 0)	F = R_N x / l - same for both directions
Drum rotation makes friction force assist (self-energising)	F = R_N (x - μ a) / l
Drum rotation opposes (friction resists the applied force)	F = R_N (x + μ a) / l
Self-locking condition	x ≤ μ a (F becomes zero or negative) -> a < x/μ must be kept

> **Block pressure**: p_b = R_N/(b l_b) ≤ allowable (0.4-1 MPa); drum wrap angle 2θ; for angle ≥ 60° the equivalent coefficient μ' = 4μ sin(θ/2)/(θ + sin θ) is used (wedge effect).

> **Heat generated** per second Hg = μ R_N v (W); block area A_b = b l_b; check p v.

> Single block loads the shaft with side thrust R_N -> use a **double block** (equal & opposite).

2. Pivoted (long) shoe external brake

> Pressure p = pa sin θ / sin θa again; same torque equation T = μ pa b r² (cos θ1 - cos θ2)/sin θa for both clutch and brake; Mn and Mf have the same form (sign of Mf reverses with direction of rotation).

> Choose the pivot position a = 4 r sin α/(2α + sin 2α) so that Mf = 0 -> uniform wear, no self-locking, same torque in both directions.

> Pivot reaction (anchor load) = vector sum of normal and friction components - needed to size the pivot pin.

Brake / clutch type	Torque equation	Notes
Internal expanding shoe	T = μ pa b r² (cos θ1 - cos θ2)/sin θa	Drum brake; centrifugal clutch
External long shoe (pivoted)	same	Railway brake, press clutch
Single block (short)	T = μ R_N r	Hand brake, small hoist
Band	T = (T1 - T2) r; T1/T2 = e^(μθ)	Winches, chain saws, auto transmission
Cone	T = μ W R_m / sin α	Synchroniser, heavy clutch
Plate / disc	T = n μ W R_m	Car clutch, disc brake: T = 2 μ F R_m

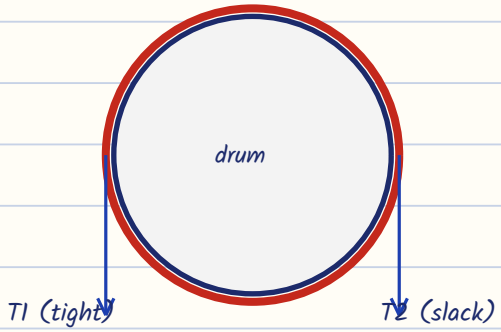
Example 2 – single block brake

> Drum 300 mm diameter, brake torque required T = 150 N·m = 150,000 N·mm, μ = 0.25, x = 250 mm, l = 600 mm, a = 40 mm (friction force offset), self-energising direction.

> $R_N = T/(\mu r) = 150{,}000/(0.25 \times 150) =$ **4000 N** ; $F = R_N (x - \mu a)/l = 4000\,(250 - 10)/600$ = **1600 N**. (If the rotation is reversed: $F = 4000 \times 260/600 = 1733$ N.)

> Self-locking if $x \le \mu a$: here 250 > 10, so not self-locking ✓.

VI. Band type clutches and brakes



$T1 / T2 = e^{(\mu \theta)}$

Brake torque $T_B = (T1 - T2)\, r$

Max band stress: $\sigma = T1 / (b\, t)$

Max pressure $p = 2\, T1 / (b\, D)$

Band lined with friction material; lever gives mechanical advantage.

> A flexible steel band (lined with friction material) is wrapped around the drum through angle θ; tightening the band produces a tension difference.

> **Euler's belt friction**: $T1 / T2 = e^{(\mu \theta)}$; brake torque $T_B = (T1 - T2)\, r$ (r = effective radius = drum radius + band thickness/2).

> Band stress $\sigma = T1/(b\, t) \le$ 80–100 MPa; maximum pressure at tight end $p_{max} = 2\, T1/(b\, D) \le 0.6$ MPa (lining).

> Wrap angles up to 270°–300°; $\mu \approx 0.25$–0.4 (lining on CI drum).

Band brake type	Lever relation (F at distance l from fulcrum)	Remarks
Simple	$F = T2\, a / l$ (slack end at a from fulcrum, tight end at fulcrum)	Self-energising in one direction, not in the other
Differential	$F\, l = T2\, b - T1\, a$ (ends on either side of fulcrum)	Self-locking if $b/a \le e^{(\mu \theta)}$; high mechanical advantage; risk of grab

Band brake type	Lever relation (F at distance l from fulcrum)	Remarks
Band and block	Wooden / lined blocks on the band; T1/T2 = ((1 + μ tan(φ/2))/(1 - μ tan(φ/2)))^n	Heavy cranes; blocks replaceable

Example 3 – simple band brake

> Drum 300 mm dia, wrap 270° (4.71 rad), μ = 0.3, required brake torque 400 N·m. e^(μθ) = e^(1.414) = 4.11.

> T1 - T2 = T/r = 400,000/150 = 2667 N ; T1 = 2667 × 4.11/3.11 = **3525 N** ; T2 = 858 N.

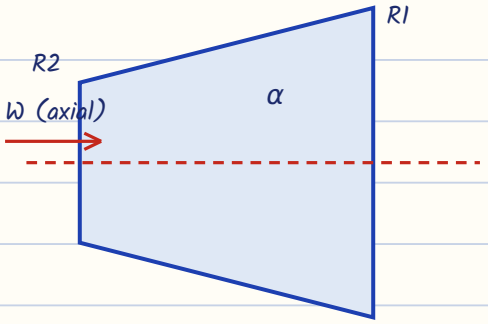
> Band width from p_max = 2 T1/(b D) = 0.6 N/mm² -> b = 2 × 3525/(0.6 × 300) = 39.2 -> **b = 40 mm**. Lever: a = 50 mm, l = 500 mm -> F = T2 a/l = 858 × 50/500 = **85.8 N**.

Band clutch

> Same theory: friction band tightened on a rotating drum connects the two shafts; torque = (T1 - T2) r. Actuated by a lever & toggle; self-energising in one direction.

> Used in earth-moving machines, winches, and **bands of automatic transmissions** (band holds the planetary drum).

VII. Cone clutches and brakes



α = semi-cone angle (10°–15°)

Rm = (R1 + R2)/2

T = μ W Rm / sin α

Engaging force F = W (sin α + μ cos α)

Cone gives a wedging action:

less axial force than a plate clutch.

Two conical surfaces (male cone, female cone) engage under axial force W. The **wedge action** multiplies the friction force (normal force N = W/(sin α + μ cos α)), so a cone clutch needs less axial force than a plate clutch of the same torque - but is bulky and may jam. Semi-cone angle α is 10°–15° (must be larger than the friction angle $\tan^{-1}\mu$ to release easily).

Case	Axial force W	Torque T
Uniform pressure p	$W = p\,\pi\,(R1^2 - R2^2)$	$T = (2/3)\,\mu\,W\,(R1^3 - R2^3) / [\,\sin\alpha\,(R1^2 - R2^2)\,]$
Uniform wear (p_{max} at R2)	$W = 2\pi\,p_{max}\,R2\,(R1 - R2)$	$T = \mu\,W\,R_m / \sin\alpha$, $R_m = (R1 + R2)/2$
Face width b & dimensions	$b = (R1 - R2)/\sin\alpha$	Normal pressure $p = W/(2\pi\,R_m\,b\sin\alpha)$
Engaging force	$F = W(\sin\alpha + \mu\cos\alpha)$ (force needed to engage while rotating)	Disengaging force $Fd = W(\mu\cos\alpha - \sin\alpha)$ (if positive)

Example 4 – cone clutch

> $R1 = 120$ mm, $R2 = 100$ mm, $\alpha = 12.5°$, $\mu = 0.25$, $T = 300$ N·m (uniform wear). $R_m = 110$ mm.

> $W = T\sin\alpha/(\mu\,R_m) = 300{,}000 \times 0.2164/(0.25 \times 110) =$ **2361 N**; face width $b = (R1 - R2)/\sin\alpha = 20/0.2164 = 92$ mm.

> Pressure $p = W/(2\pi\,R_m\,b\sin\alpha) = 2361/(2\pi \times 110 \times 92 \times 0.2164) = 0.17$ N/mm² ✓; engaging force $F = W(\sin\alpha + \mu\cos\alpha) = 2361\,(0.2164 + 0.244) =$ **1087 N**.

Single plate clutch and disc brake (automobile) – quick relations

> **Single plate clutch**: two friction faces ($n = 2$) lined with organic material: $T = 2\,\mu\,W\,R_m$ (uniform wear). Spring force W = (6-9 springs). Capacity = 1.3 × engine torque.

> **Disc brake**: caliper squeezes two pads on a rotor: $T = 2\,\mu\,F\,R_m$ (F = clamp force each side, R_m = mean pad radius); better heat dissipation & no self-energising (stable, less fade) vs drum brake.

> Drum vs disc: drum is self-energising (high gain, cheap, parking brake) but fades & heavy; disc has uniform torque, ventilated rotor, quick recovery.

VIII. Energy considerations

All the energy of a brake (and the slip energy of a clutch) is converted into **heat**. The brake or clutch must be designed so that this heat does not overheat the friction surfaces.

Case	Energy absorbed E (J)
Rotating body slowed from ω1 to ω2	$E = \tfrac{1}{2}\,I\,(\omega1^2 - \omega2^2)$, I = moment of inertia (kg m²)
Translating body (vehicle) from v1 to v2	$E = \tfrac{1}{2}\,m\,(v1^2 - v2^2)$

Case	Energy absorbed E (J)
Lowered load (hoist) through height h at constant speed	E = m g h (potential energy) + kinetic part
Combined	E = ½ I (ω1² - ω2²) + ½ m (v1² - v2²) + m g h (+ work done by friction etc.)
Brake torque, time	T = I (ω1 - ω2) / t (constant deceleration); stopping angle θ = ω1 t/2 ; E = T θ
Clutch engagement (driven inertia I2 from rest to ω1 with constant torque T)	Slip time t = I2 ω1 / T ; energy dissipated = ½ I2 ω1² (equals the KE gained by the driven part); mean power = E/t

> **Equivalent inertia** of a vehicle at the wheel: I_eq = m r_w² + rotating parts; for brakes on 'n' wheels the energy is shared according to the weight transfer (front 60-70%).

> Frequent clutch operation (starting) should have short slip time but also gentle engagement: engagement time 0.5-1 s for cars; greater energy for heavy trucks & large inertia.

IX. Temperature rise

$$\Delta t = E / (m\ c) \quad \text{(single stop)} \quad ; \quad \text{steady: } Hg = \mu\ p\ v\ A = Hd = h\ A\ (t - t0)$$

> m = mass of the parts that absorb heat (drum/ rotor, kg); c = specific heat (cast iron / steel ≈ 460-500 J/kg°C).

> **Single application**: all heat absorbed by the drum (short time, no loss): Δt = E/(m c). Limit Δt ≈ 100 °C for single stop; surface temperature ≤ 250-350 °C (drum), ≤ 600 °C (disc).

> **Continuous (cyclic) braking**: heat generated per second Hg = Ft v = μ p A v ; heat dissipated Hd = h A (t - t0) with h = 15-30 W/m²°C (still air) to 60-100 W/m²°C (ventilated, moving vehicle) -> equilibrium temperature rise t - t0 = μ p v / h.

> **p v rule**: the product p v (MPa × m/s) limits wear and temperature: for organic lining p v ≤ about 1-2 MPa m/s (dry); in oil 2-3.

> Cooling: ribs/fins on the drum, ventilated (drilled / vane) disc rotors, air ducts, oil spray (wet clutch), large drums with high thermal mass; avoid brake fade (loss of μ at high temp) by choosing materials.

Example 5 – energy and temperature rise in a vehicle brake

> Car of mass 1200 kg stopped from 80 km/h (22.2 m/s). $E = \frac{1}{2} m v^2 = 0.5 \times 1200 \times 22.2^2 =$ **296 kJ**. Four brakes (equal): 74 kJ each (front brakes actually take 60–70%).

> Each disc/ drum has m = 6 kg of cast iron, c = 500 J/kg°C: $\Delta t = 74{,}000/(6 \times 500) =$ **24.7 °C** per stop (rises sharply with repeated stops: 5 stops in succession -> about 120 °C if no cooling).

> Time to stop with deceleration 0.6 g = 5.9 m/s²: t = 22.2/5.9 = 3.8 s ; mean braking power = 296 kJ/3.8 s = 78 kW (all 4 brakes) - therefore good cooling is vital.

Example 6 – clutch slip energy

> Driven inertia I2 = 0.8 kg m² picked up from rest to engine speed 2000 rpm (209 rad/s) by clutch torque 120 N·m.

> Slip time $t = I2\,\omega I/T = 0.8 \times 209/120 = 1.4$ s ; energy dissipated $E = \frac{1}{2}\, I2\, \omega I^2 = 0.5 \times 0.8 \times 209^2 =$ **17.5 kJ** ; mean heating rate = 17.5/1.4 = 12.5 kW.

> If the pressure plate + flywheel face mass is 8 kg (c = 500): $\Delta t = 17{,}500/(8 \times 500) =$ **4.4 °C** - fine for one start; repeated starts need checking.

X. Quick revision & important questions

Formulas to remember

> Pressure: uniform pressure $T = (2/3)\,\mu\, W\, (R1^3 - R2^3)/(R1^2 - R2^2)$; uniform wear $T = \mu\, W\, R_m$ ($R_m = (R1 + R2)/2$).

> Shoe: $T = \mu\, pa\, b\, r^2\, (\cos \theta 1 - \cos \theta 2)/\sin \theta a$; $F = (Mn \mp Mf)/c$; self-locking if $Mn \le Mf$.

> Block: $F = R_N\, (x \mp \mu\, a)/l$; $T = \mu\, R_N\, r$. Band: $T1/T2 = e^{(\mu\theta)}$; $T = (T1 - T2)\, r$; $F = T2\, a/l$ (simple).

> Cone: $T = \mu\, W\, R_m / \sin \alpha$; $F = W\, (\sin \alpha + \mu \cos \alpha)$. Energy: $E = \frac{1}{2}\, I\, (\omega 1^2 - \omega 2^2) = \frac{1}{2}\, m\, v^2$; $\Delta t = E/(m\, c)$.

1. Compare clutches and brakes; list the friction materials with μ and p_{max}. (5 M)
2. Derive the torque capacity of a clutch for uniform pressure and uniform wear. (10 M)

3. Analyse an internal expanding shoe brake: torque, actuating force and self-locking. (10 M)
4. Explain the external contracting shoe brake: single block and pivoted long shoe. (10 M)
5. Derive the relation for a band brake; explain simple and differential band brakes. (10 M)
6. Design a cone clutch to transmit given power (find dimensions, axial force). (10 M)
7. Explain energy considerations and temperature rise in clutches/ brakes. (5 M)
8. A vehicle brake absorbs given energy - find the temperature rise. (5 M)